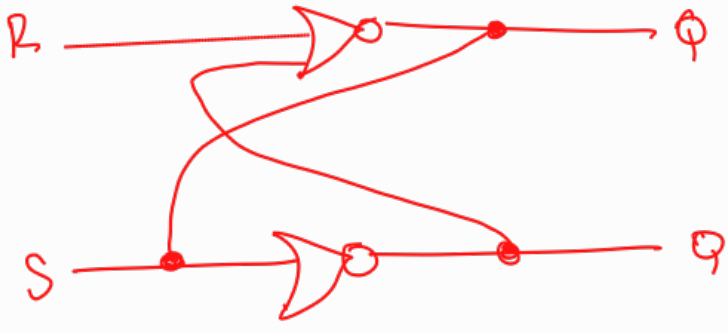
combinational vs sequential circuits

combinational → consists of logic diagrams / input variables and one output.

sequential circuits → consist of storage elements.
output depends on input and currently stored value in storage

State → value currently stored in memory.

filp flops are made with the help of latches. and these filp flops are the major element when making storage

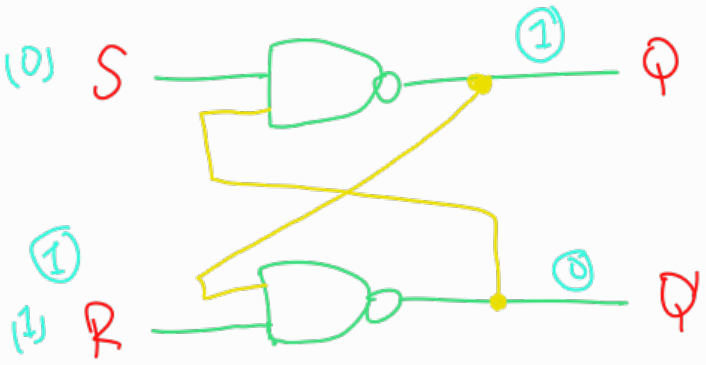


Set → save 1 at Q

Reset → save 0 at Q

the rest is complement

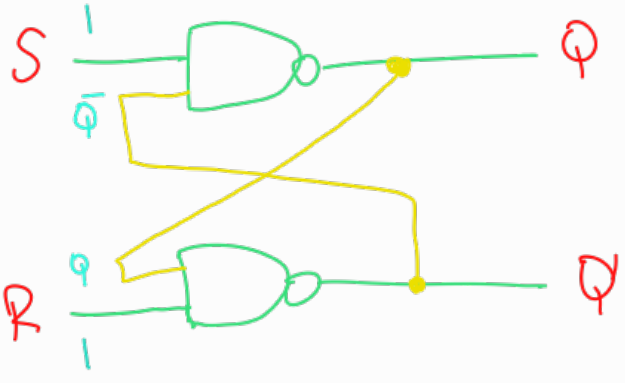
S R latch with nand gate



A	B	F
0	0	1
0	1	1
1	0	1
1	1	0

S	R	Q	Q'
0	0	Invalid	Invalid
0	1	1	0
1	0	0	1
1	1	No·C	No·C → Previous State

Whenever one of the inputs is zero the Ans is always 1.



$S = \overline{1 \cdot \bar{Q}}$ $\quad S = \bar{1} + \bar{\bar{Q}}$

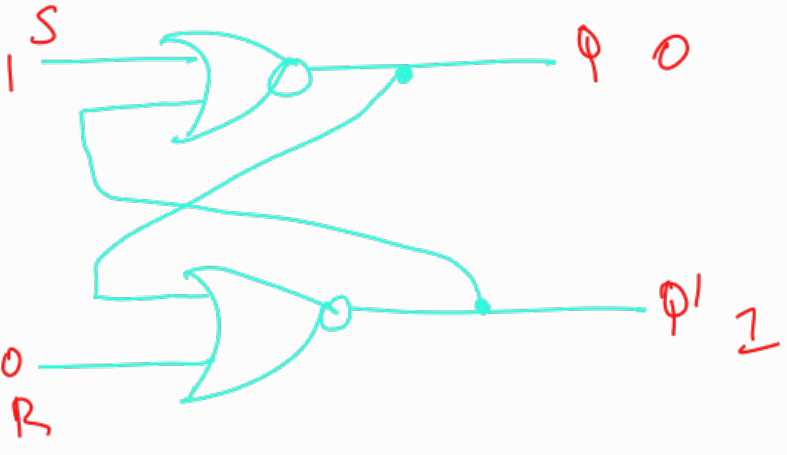
$S = 0 + Q = Q$

$R = \overline{1 \cdot Q}$ $\quad R = \bar{1} + \bar{Q}$

$= 0 + \bar{Q} = \bar{Q}$

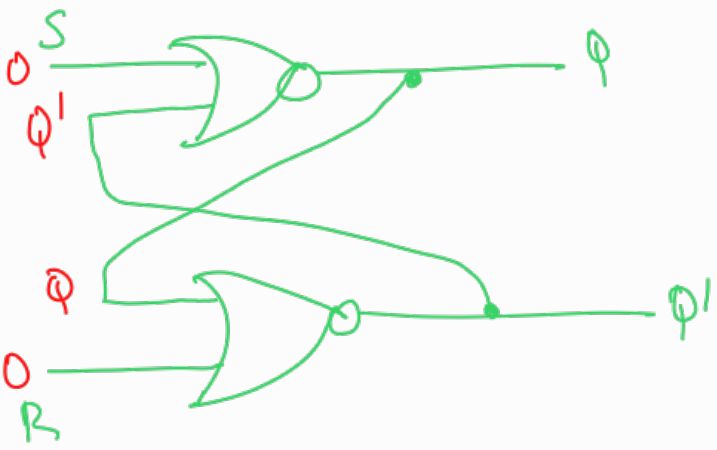
hence when both S and R are 1 we get same values

S R latch using Nor gates



A	B	F
0	0	1
0	1	0
1	0	0
1	1	0

if one of the Input is 1 then output is zero.



S	R	Q	Q'
0	0	Same	Same
0	1	1	0
1	0	0	1
1	1	invalid	Invalid

$S = \overline{0 + Q'}$ $\quad R = \overline{0 + Q}$

$S = \bar{0} \cdot \bar{\bar{Q}}$ $\quad R = \bar{0} \cdot \bar{Q}$

$S = 1 \cdot Q = Q$ $\quad R = 1 \cdot \bar{Q} = \bar{Q}$

gives Same output

$Q(t)$ → current State $\quad$ $Q(t+1)$ → Next State

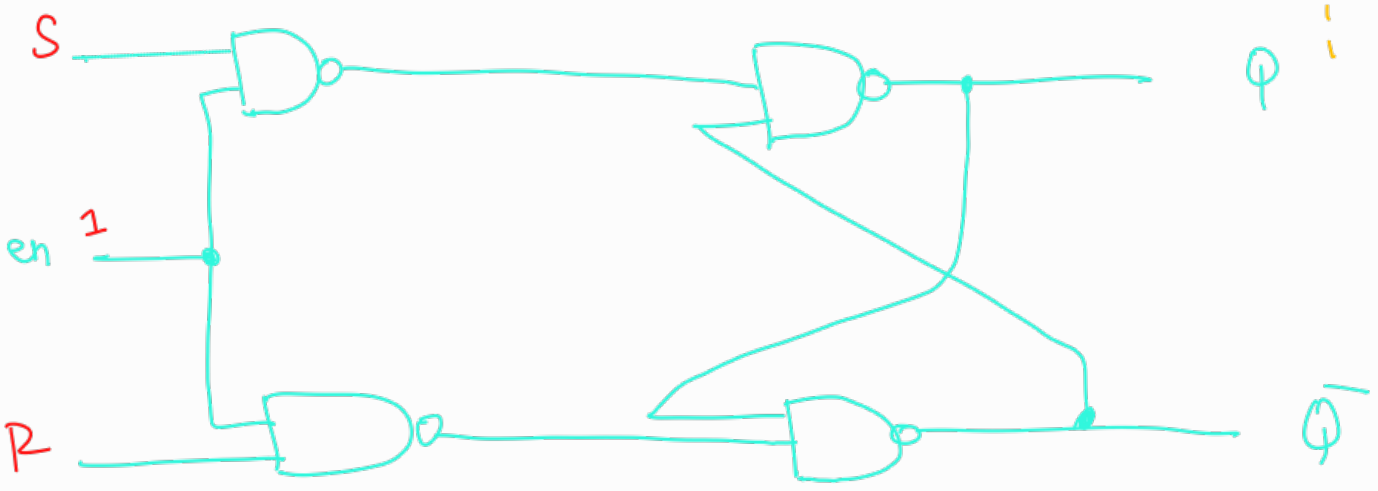
note Q output is 1 then latch is at Set position

Q output is 0 then Latch is at reset position.

S	R	Q	Q'	
0	0	Same	Same	
0	1	1	0	Set
1	0	0	1	reset
1	1	invalid	Invalid	

SR latch with control input (enable input) a.k.a gated Latch.

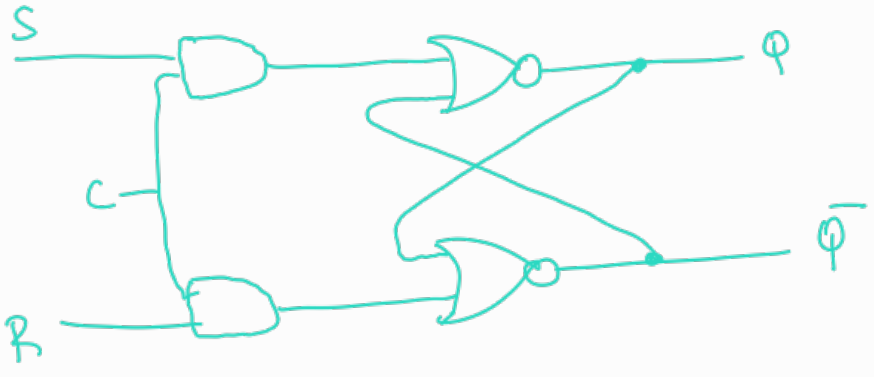
A	B	F
0	0	1
0	1	1
1	0	1
1	1	0



En	S	R	Q	Q'	
0	X	X	no change	no change	→ same state
1	0	0	same	same	
1	0	1	0	1	→ Reset
1	1	0	1	0	→ Set
1	1	1	undefined	undefined	

Latches never generate a 0 output because of memory

SR latch with control input



D- Latch a.k.a

data Seeector a.k.a

transparent Latch

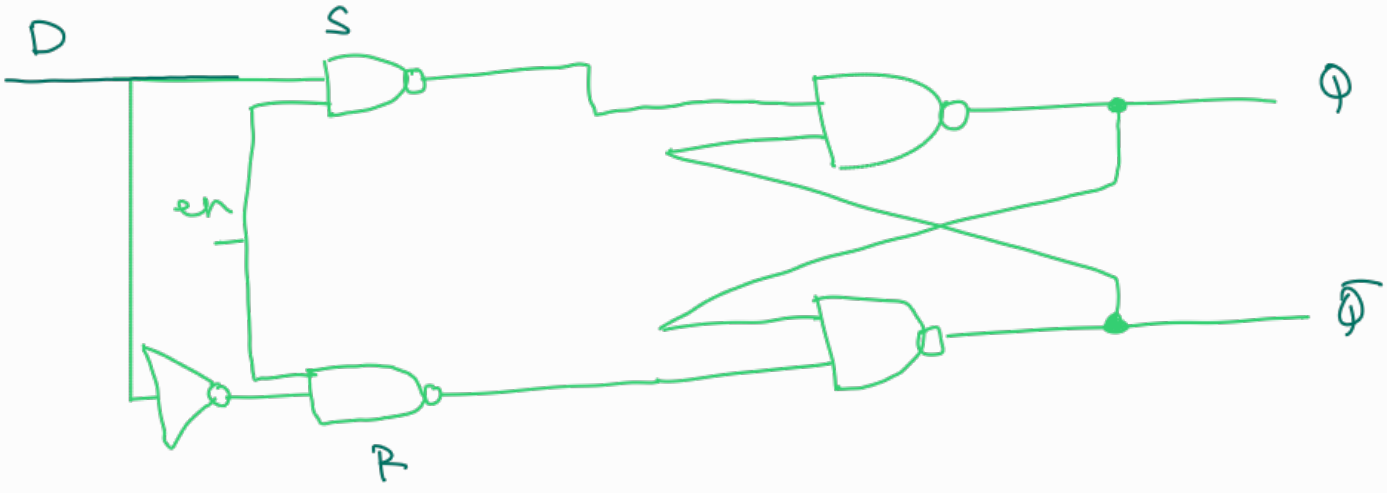
gives us output only for Set and resets State. It ignores all other Undefined and no change states.

Set $Q \rightarrow 1$

Reset $Q \rightarrow 0$

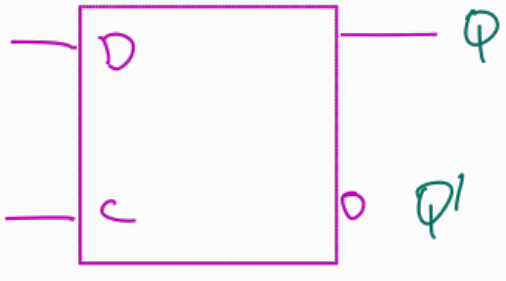
Set and reset states occur when S and R latches are complements of each others.

E	S	R	Q	Q'	
1	0	1	0	1	(Reset)
1	1	0	1	0	(Set)

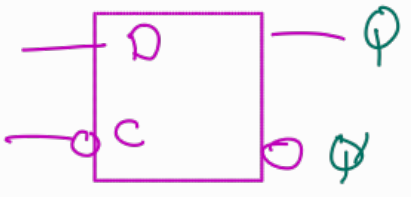


En	D	Q	Q'	
0	X	no c	no c	
1	0	0	1	Reset
1	1	1	0	Set

graphical Symbol



D with 1 enable/control input

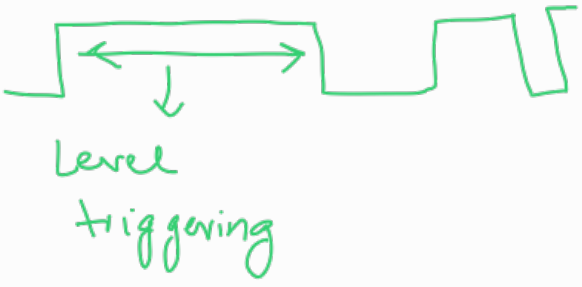


D with 0 control input

(Active high) (Active Low)

Difference between Latches and filp flop.

Latches	filp flops
→ Bulding blocks of sequential circuits	Bulit from latches.
→ Based on enable input	works on the basis of clock pulses.
→ level triggered	→ edge triggered



process is controlled by a clock

filpflop = Symmetric and Synchronised.

enable → (user dependent)

clock → Synchronised Single generator.

olp depends on the ip and the previous state which is stored in the memory.

The memory element changes from one state to an other by help of clock. (memory is triggered by clock pulse). This means it depends on clock for change in state

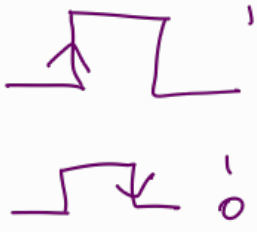
clock → converts latches to ff

→ for synchronous operation, clock is necessary

→ O/ps does not change immediately when inputs change unlike Latches.

I) Positive edge triggered

II) negative edge triggered

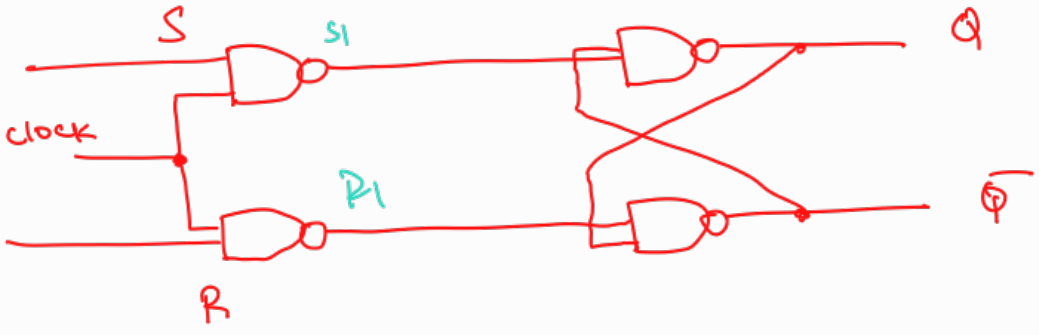


> at these stages output in ff changes.

clock: decides the time of input and generate the output.

note: clock cycle time taken when zero = time taken when 1

SR filpflop



clock has replaced the enable input.

equations at $S1 = \overline{S \cdot clock} = \bar{S} + \overline{clock}$

at $R1 = \overline{R \cdot clock} = \bar{R} + \overline{clock}$

CASE1 if clock = 0

$S_1 = \overline{S} + \overline{0} = \overline{S} + 1 = \boxed{1}$

$R_1 = \overline{R} + \overline{0} = \overline{R} + 1 = \boxed{1}$

Hence Both S_1 and R_1 are now 1 and are the inputs of Nand latch. So Q and $\overline{Q}$ have the same values. / whenever CLK is Low it will always give same output. Regardless of S_1 / R_1

CASE2

when CLK = 1

$S_1 = \overline{S} + \overline{CLK}$

$S_1 = \overline{S} + \overline{1}$

$\boxed{S_1 = \overline{S}}$

$R_1 = \overline{R} + \overline{CLK}$

$R_1 = \overline{R} + \overline{1}$

$\boxed{R_1 = \overline{R}}$

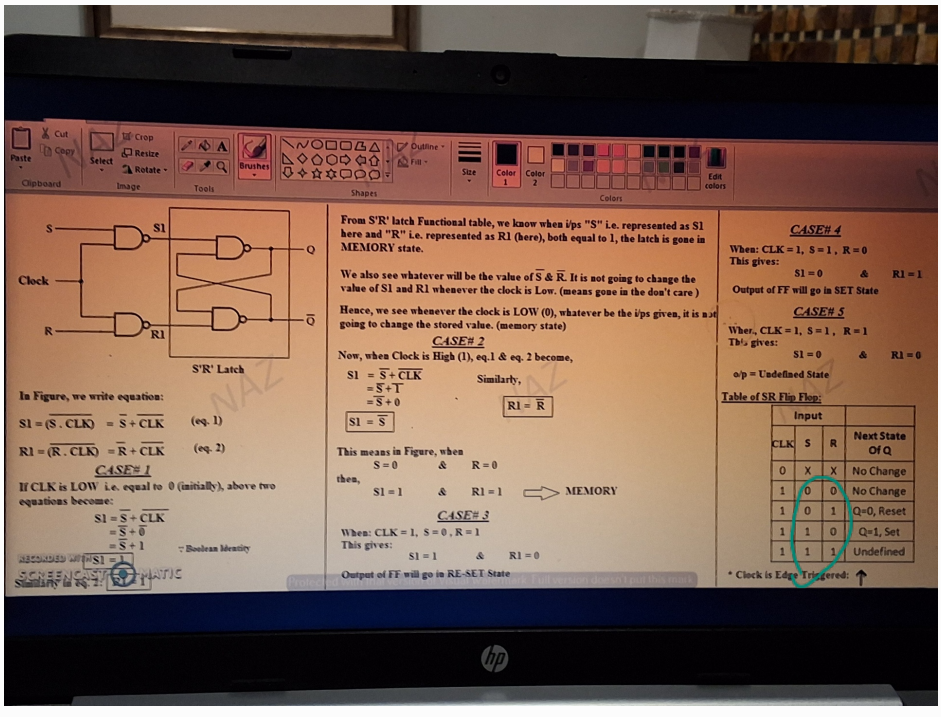
Now giving the inputs. $\dfrac{S=0 \qquad R=0}{S_1=1 \qquad R_1=1}$

CLK	S	R	Q
0	X	X	Same
1	0	0	Same

CASE 3 CLK = 1 S = 0 and R = 1

S1 = 1 and R1 = 0

CLK	S	R	Q
1	0	1	Q = 0 Reset



Make two equations

→ when clock is low

→ when clock is high

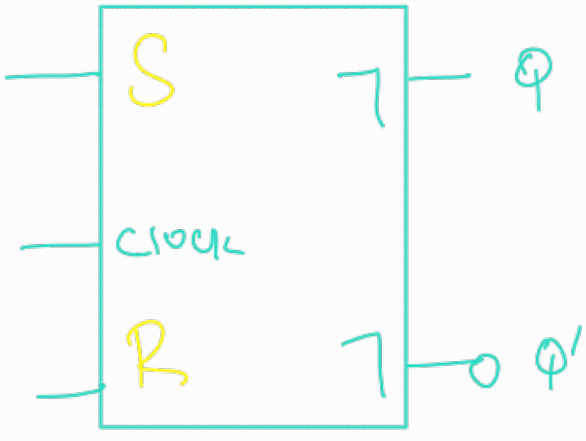
2 inputs hence $2^2 = 4$

Transparency issues with latches:

outputs suddenly changes as soon the input changes and can cause a chain reaction. and is hard to control.

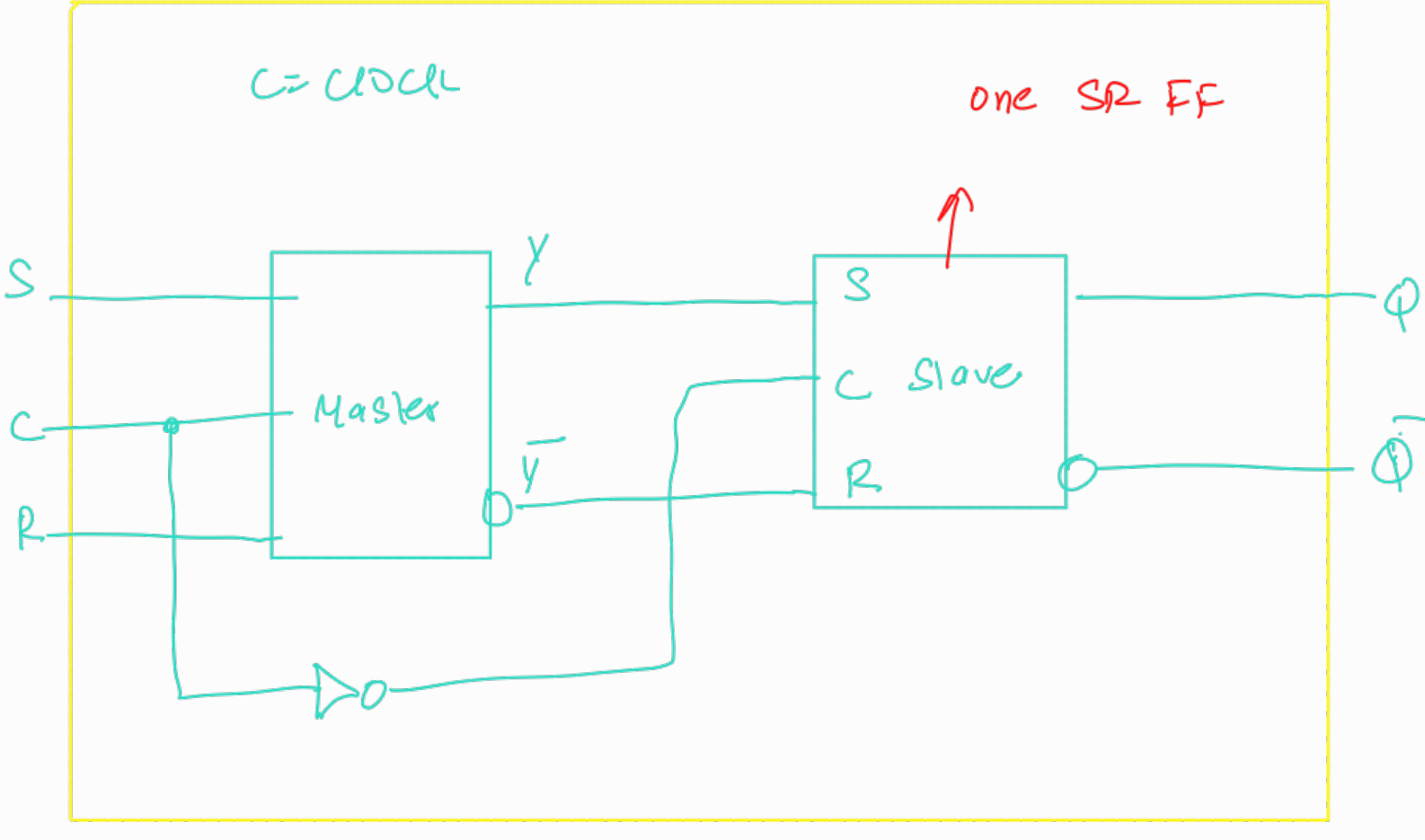
SR Master Slave FF

↗ → status update



→ diagram of SR FF

diagram for Master Slave SR FF



When C=0 then Master is disabled and Slave is enabled.

↳ means maintain previous value

When C=1 then Master is enabled and Slave is disabled.

SR 6 → video

↳ meaning maintains the previous value

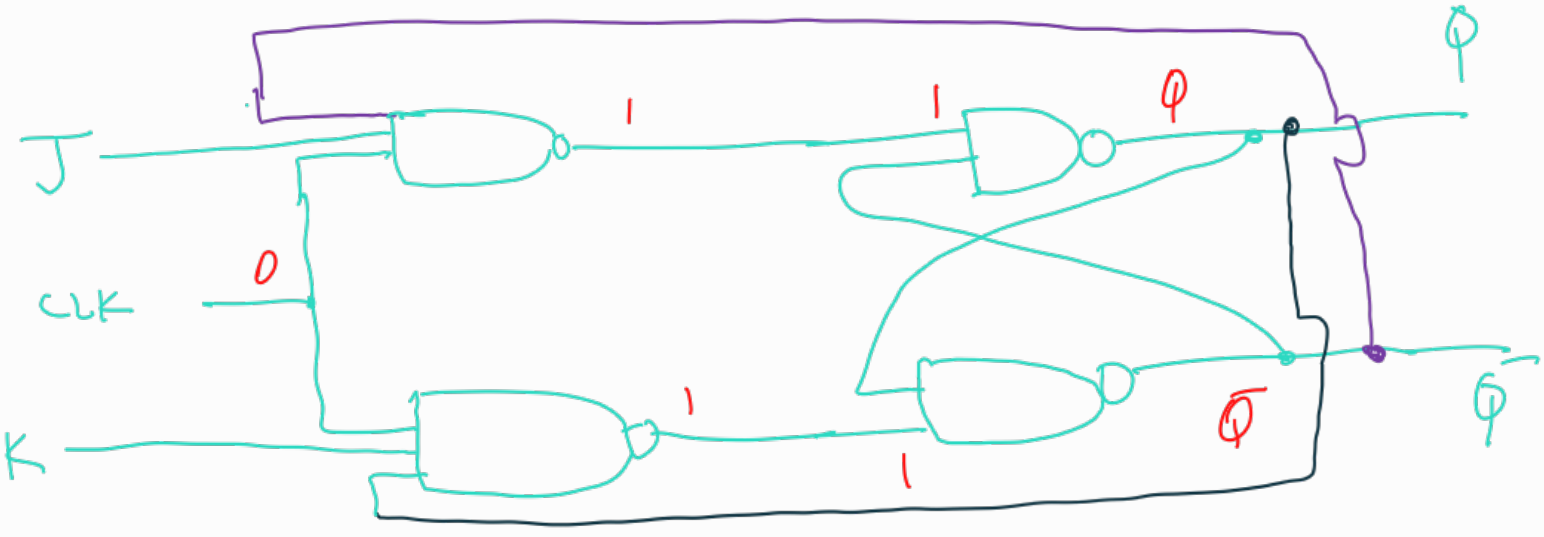
Master Slave

→ Are always edge-triggered

→ ensures stable output

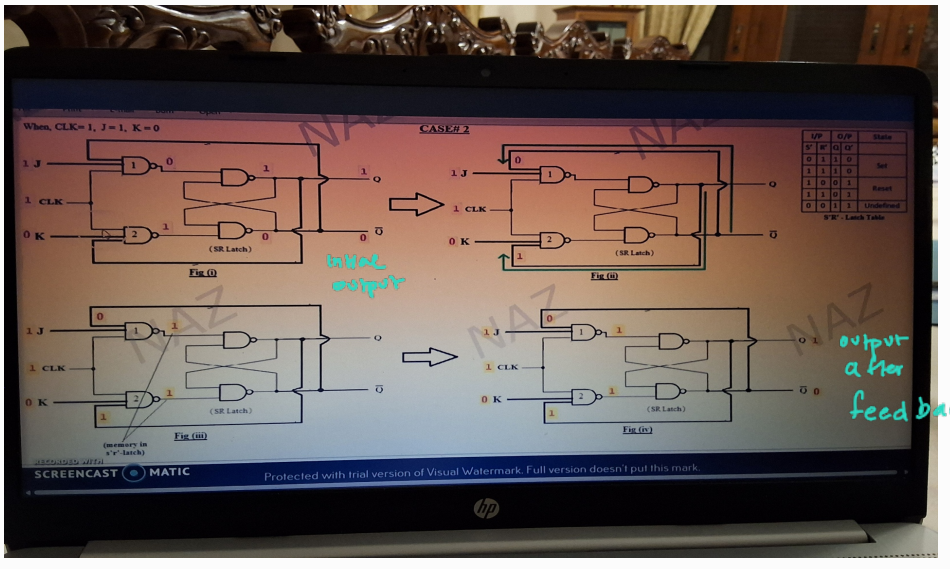
→ Prevents glitches / race around conditions

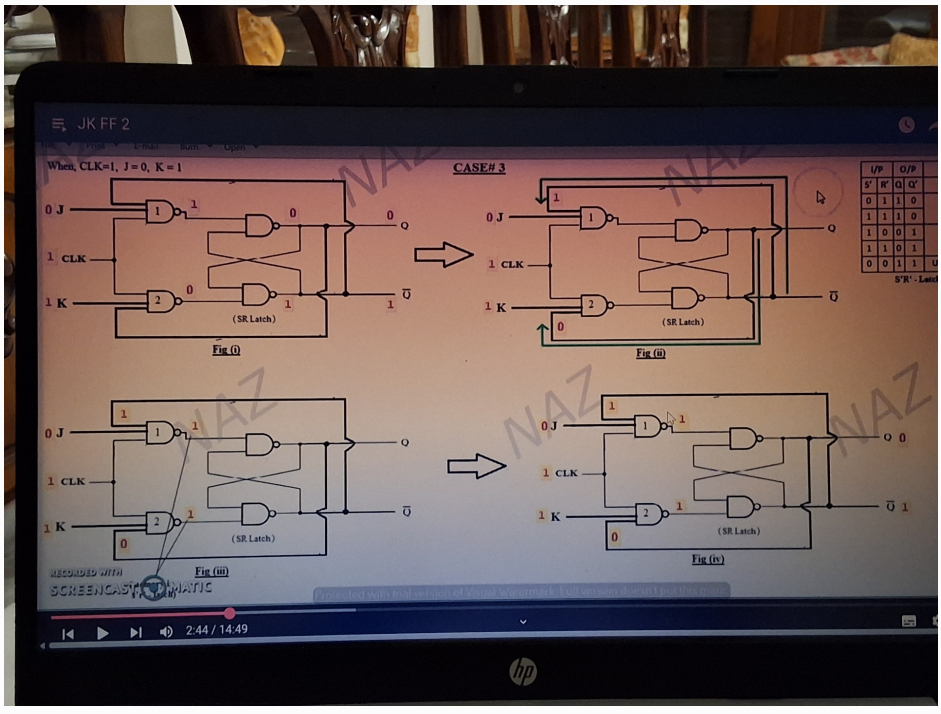
JK flipflop → Solves the invalid case for SR.

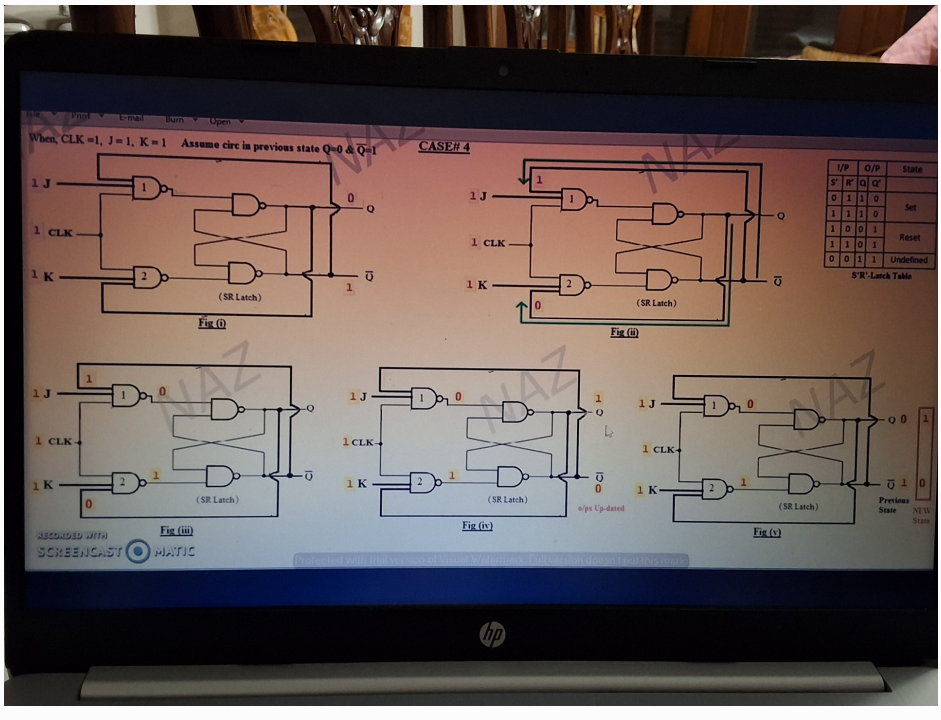


JK ff is derived from SR FF

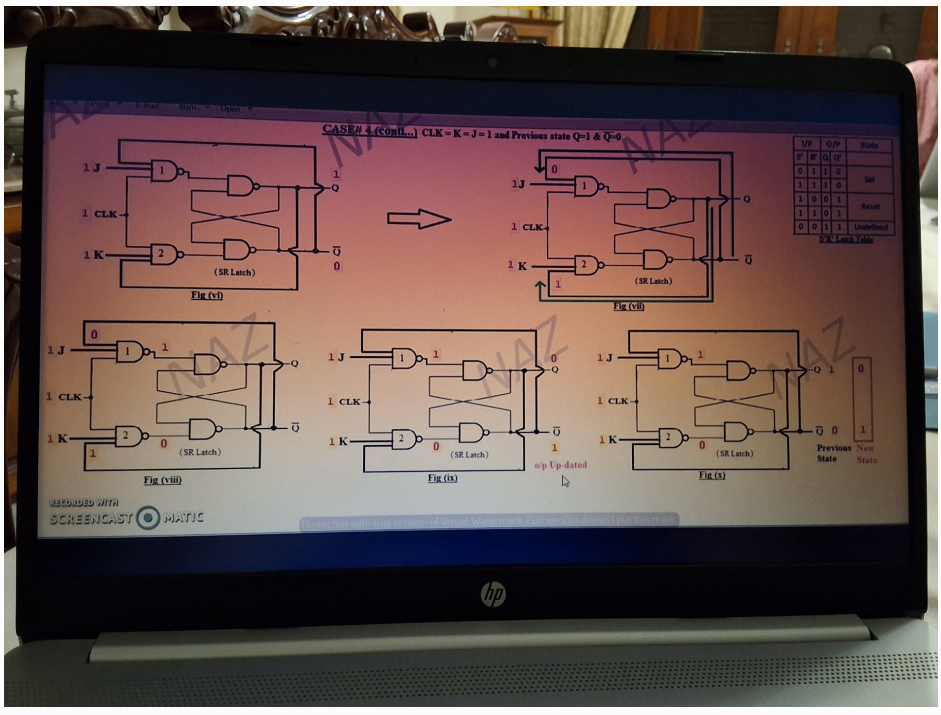
case I : CLK = 0. If any one input for nand is 0 Answer will always be equal to 1. Hence it will go to Memory state (Same values for Q and Q̄.





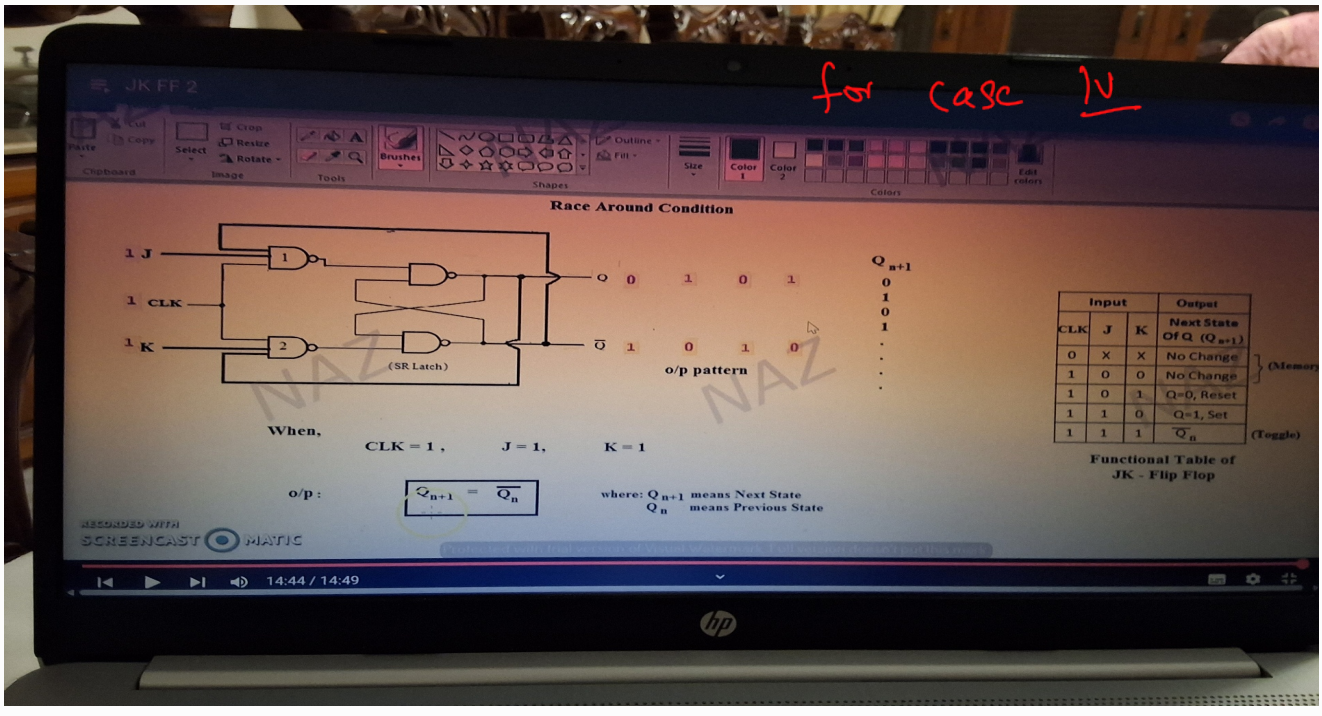


Note for J=1 K=1 and CLK=1 we will assume that previous state output $Q = 0$ and $\bar{Q} = 1$ exist firstly.



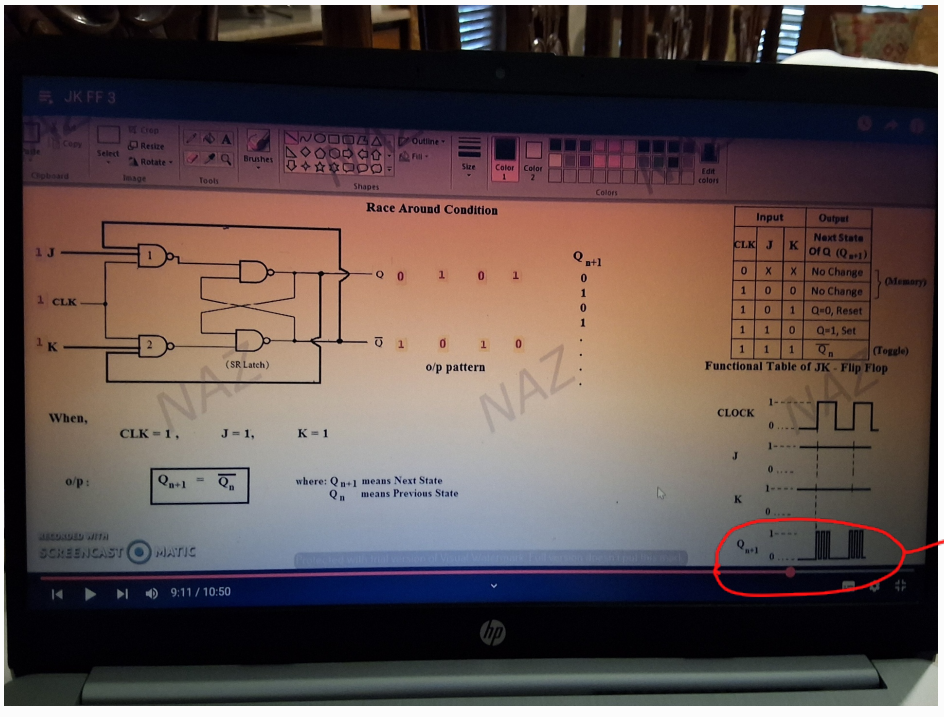
Again updates the output values after feedback

In JK ff whenever the value is 1 and 1 the output changes continuously | complement of each Previous State /toggle



whenever our J=1 and K=1 we are getting complement of our previous state

Using a Master Slave JK ff it will eliminate the toggle. Meaning output of normal JK ff varying from 0 → 1 and 1 → 0. This is a.k.a Race around condition.



JK ff actually is same as compared to SR ff. only for case (11) it changes the output from invalid (in case of SR) to toggle.

→ toggle.

note:.

Characterists table for SR ff.

S	R	Q(t)	Q (t+1)
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

↳ Previous State

→ Characteristic table with three inputs.

Since we know from SR FF T.T

S	R	Q	Q'
0	0	Same	Same
0	1	0	1
1	0	1	0
1	1	Invalid	Invalid

Previous ↓ next ↓

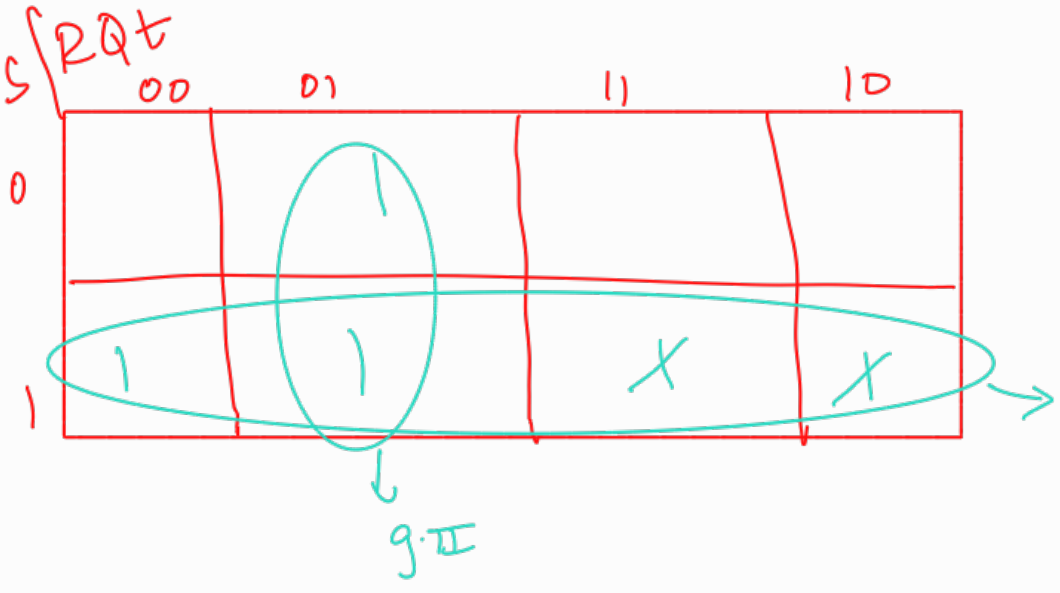
S	R	Q(t)	Q (t+1)
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	X
1	1	1	X

Invalid

filling this now

Characterist equation:.

By making K-map $2^{\text{Inputs}} \rightarrow 2^3 = 8$ K-map



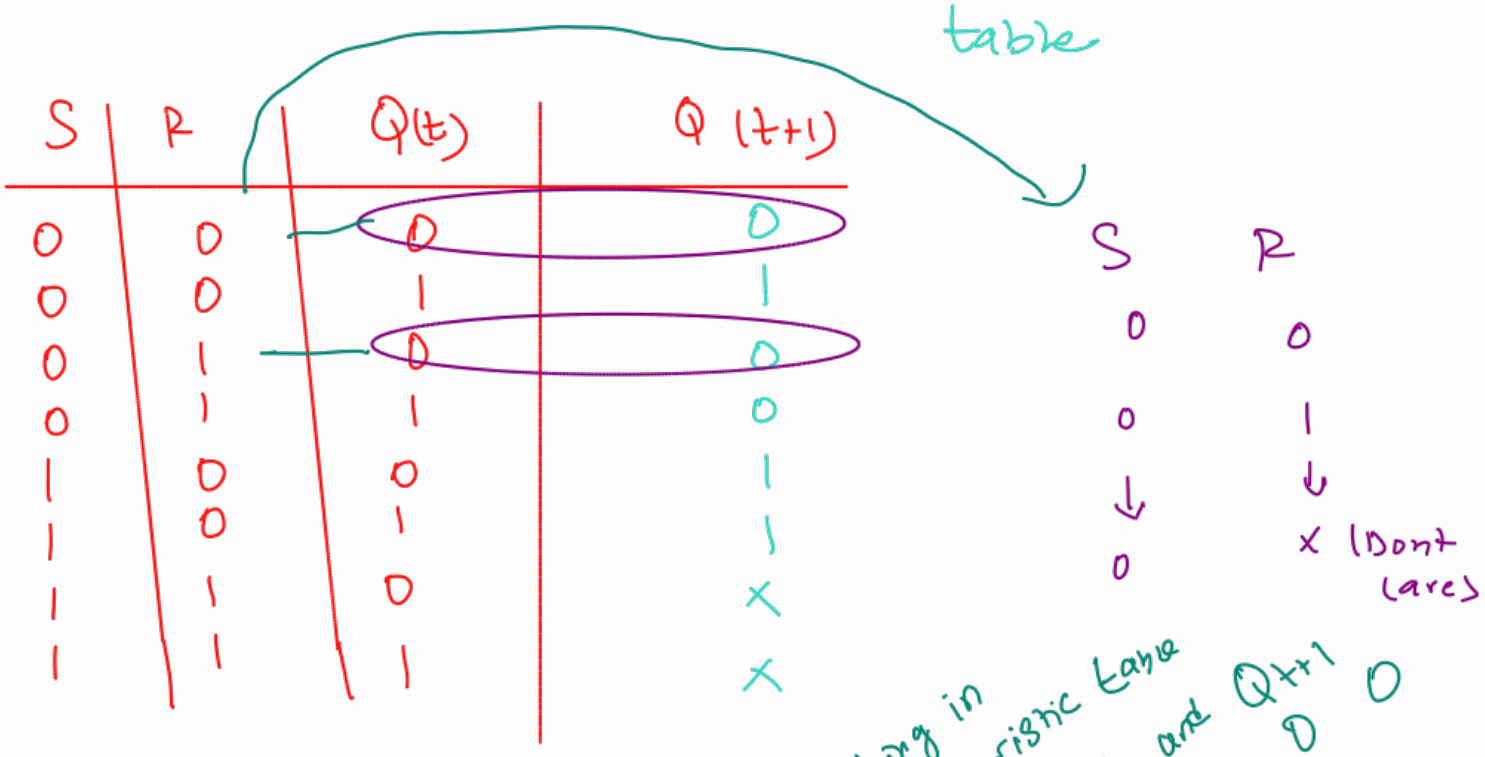
filling where Q(t+1) is 1

SOP

g.I $S + \bar{R} Qt$

$$Q_{(t+1)} = S + \overline{R}\, Q_t$$

Excitation table:- It is made from characteristic table



Looking in characteristic table where Qt and Qt+1 value is 0 0

excitation table

Q(t)	Q(t+1)	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

In excitation table we will fill the inputs by looking at where the Q(t) and Q (t+1) values are 0. (case one). Look from the characteristic table

JK FF characteristic table and excitation table

JK T·T

J	k	Q_t	$\overline{Q}_t$
0	0	Same	Same
0	1	0	1
1	0	1	0
1	1	Toggle	Toggle

(Characteristic equations)

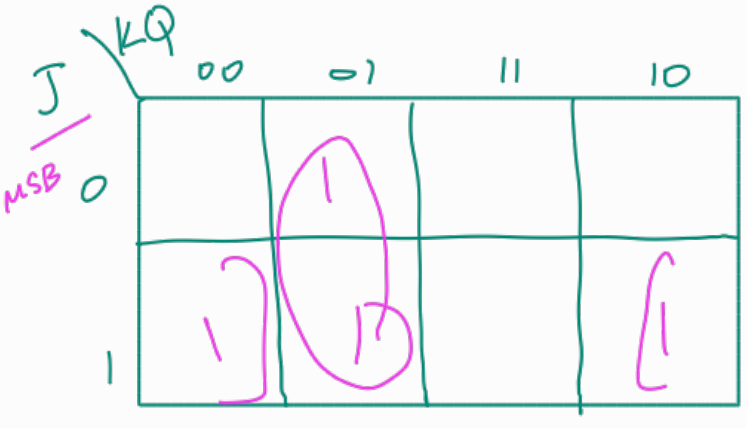
characteristic table.

J	k	Q_t (Previous)	Q_{t+1} (next)
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

inputs

toggled

Previous state



$\overline{K}Q + J\overline{Q}$

Excitation table

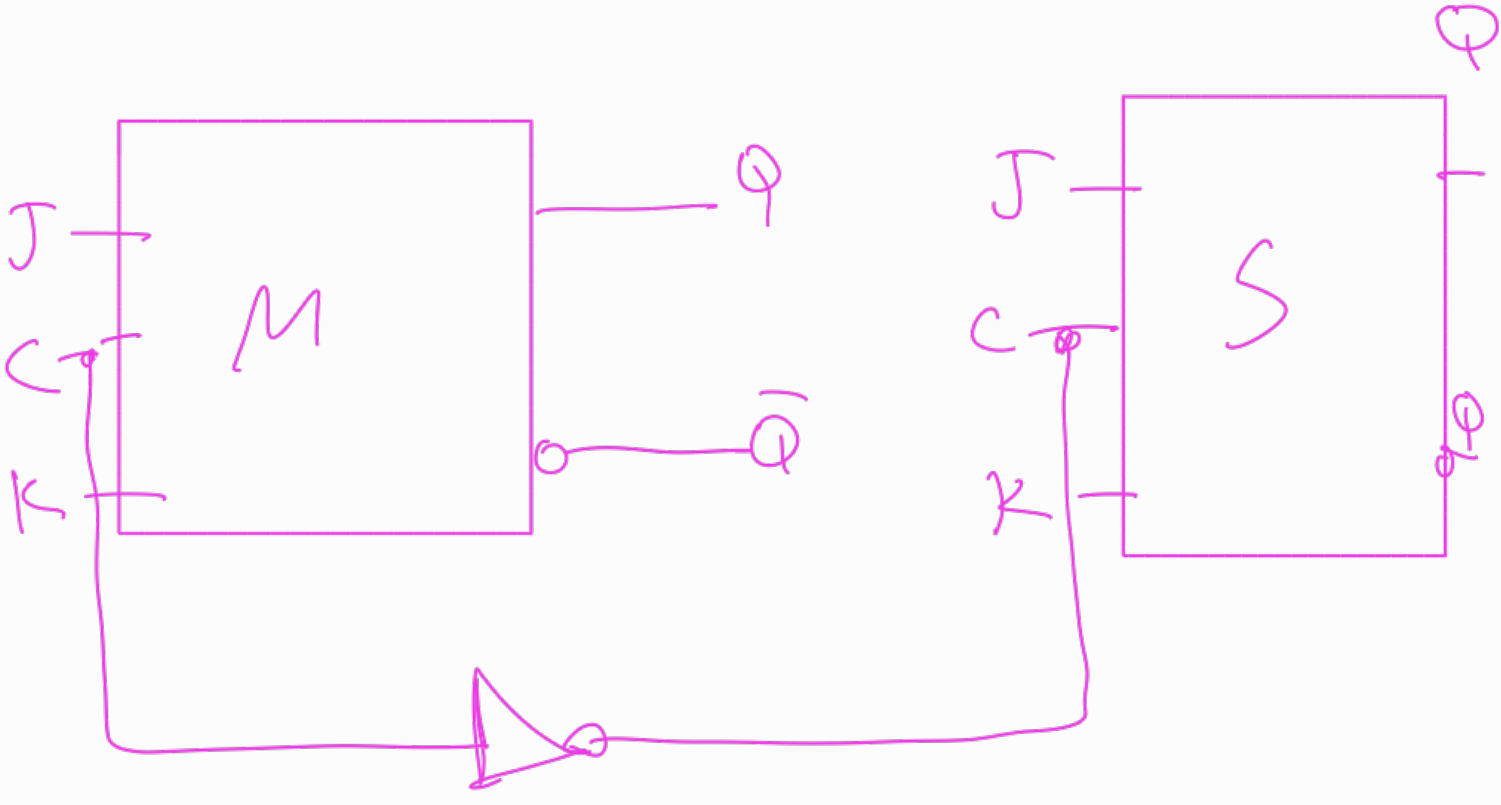
Q	$Q_{(n+1)}$	J	K
0	0	0	(X)
0	1	1	(X)
1	0	X	1
1	1	X	0

characteristic table.

J	K	Q_t	Q_{t+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

J	K
0	0
0	1
	(X)

Master Slave JK FF (Used to Solve toggle)



D filpflop working

En	D	Q	Q'
0	X	no·c	no·c
1	0	0	1
1	1	1	0

Reset

Set

Input

giving same output as input

characteristic table

D	Q_n	Q_{n+1}
0	0	0
0	1	0
1	0	1
1	1	1

2 inputs

Characteristic equation

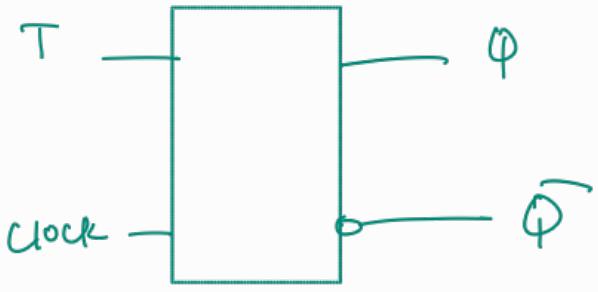
$$Q_{n+1} = D$$

Excitation table

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

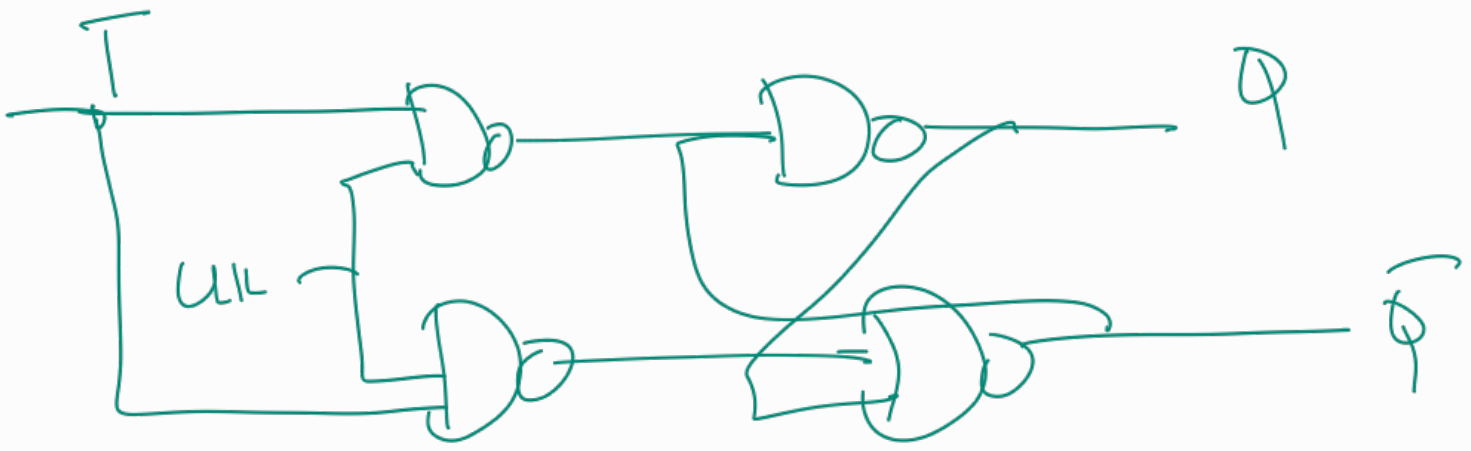
T filpflop

Toggle



Truth table

T	Q_{n+1}
0	Q_n
1	$\bar{Q_n}$



T	Q_{n+1}
0	Q_n
1	$\bar{Q}_n$

=

1 Input only

if Input is 0 then output is current state

Characteristic table

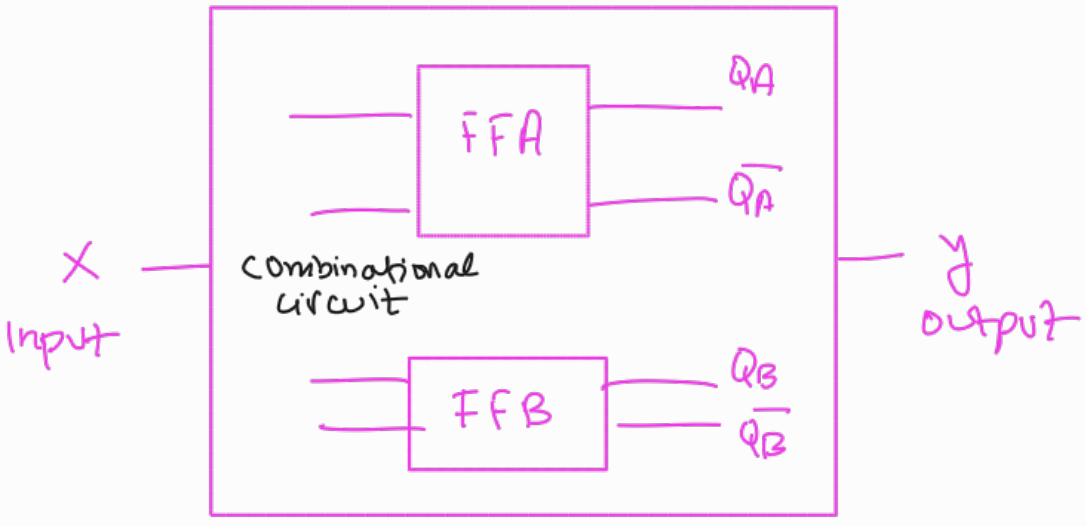
T	Q_n	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

$Q_{n+1} = T \oplus Q_n$

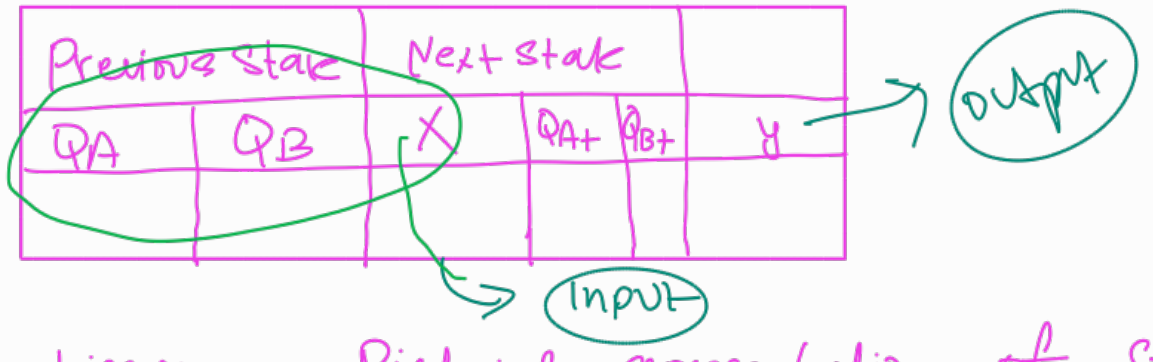
Excitation table

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

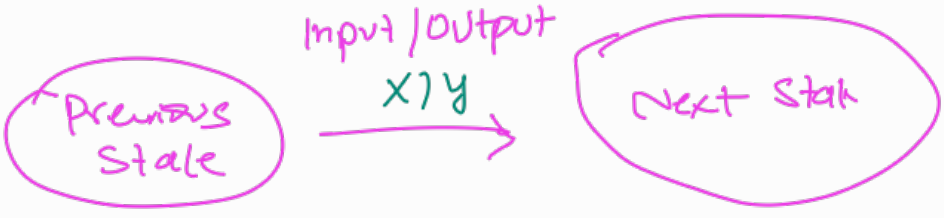
Sequential circuit Analysis.



State table :: It tells about the relation b/w Previous State, Next state, output



State diagram :: Pictorial representation of State table



Deciding total no of States :-

$2^2 = 4$

$2^{\text{no of FF}} = \text{no of States}$

Assume

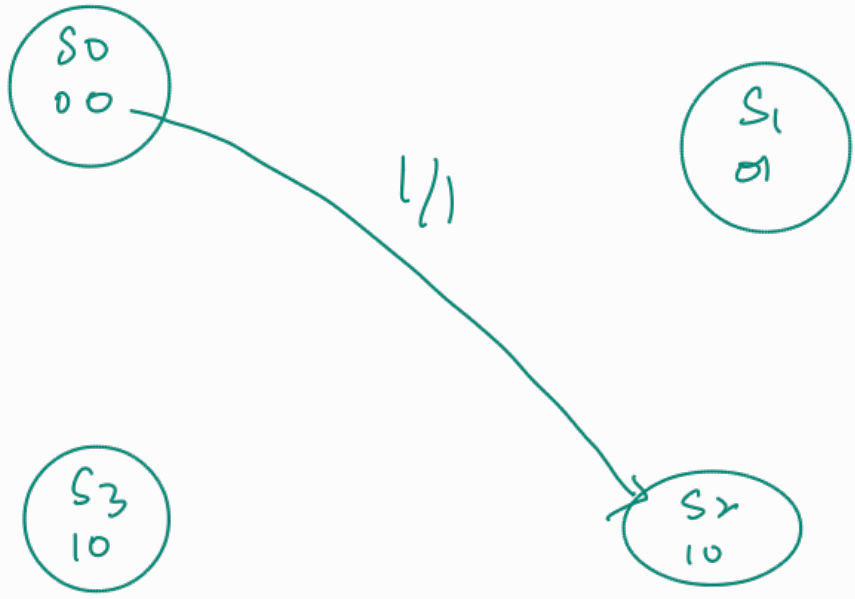
Previous		Next			output
Q_A	Q_B	x	Q_A^+	Q_B^+	y
0	0	1	1	0	1

generalic

output

Input

Drawing state diagram with help of State table



S0 → naming State.

numbering is clockwise to avoid extra mess.

Writing S0 → Binary equivalent in circle

Input / output

Previous ⟶ next

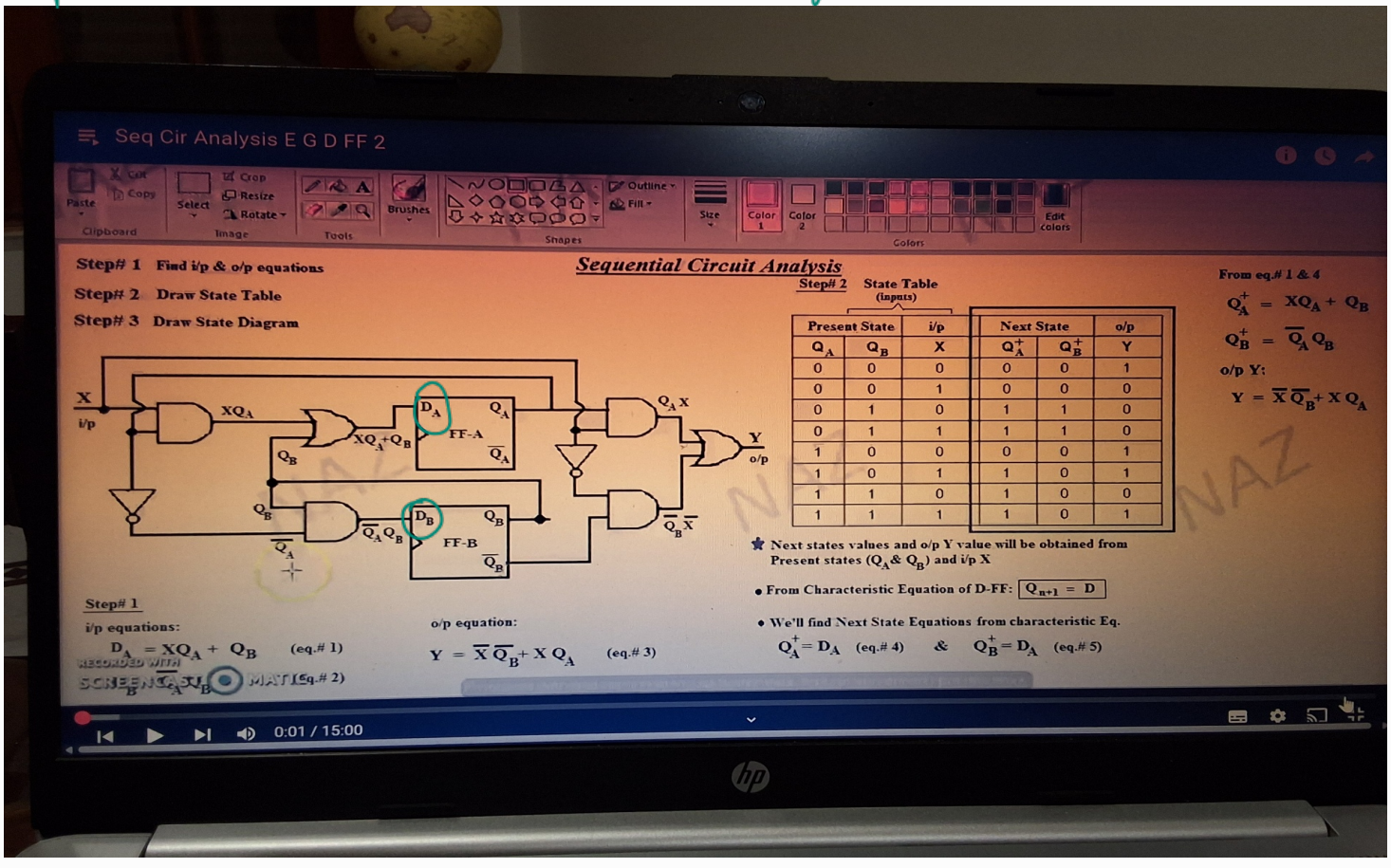
Previous		Next			output
Q_A	Q_B	x	Q_A^+	Q_B^+	y
0	0	1	1	0	1

Question:- to Solve Sequential circuit

Step 1 find the input equation of ff. and output equations.

Step 2 Draw State Table

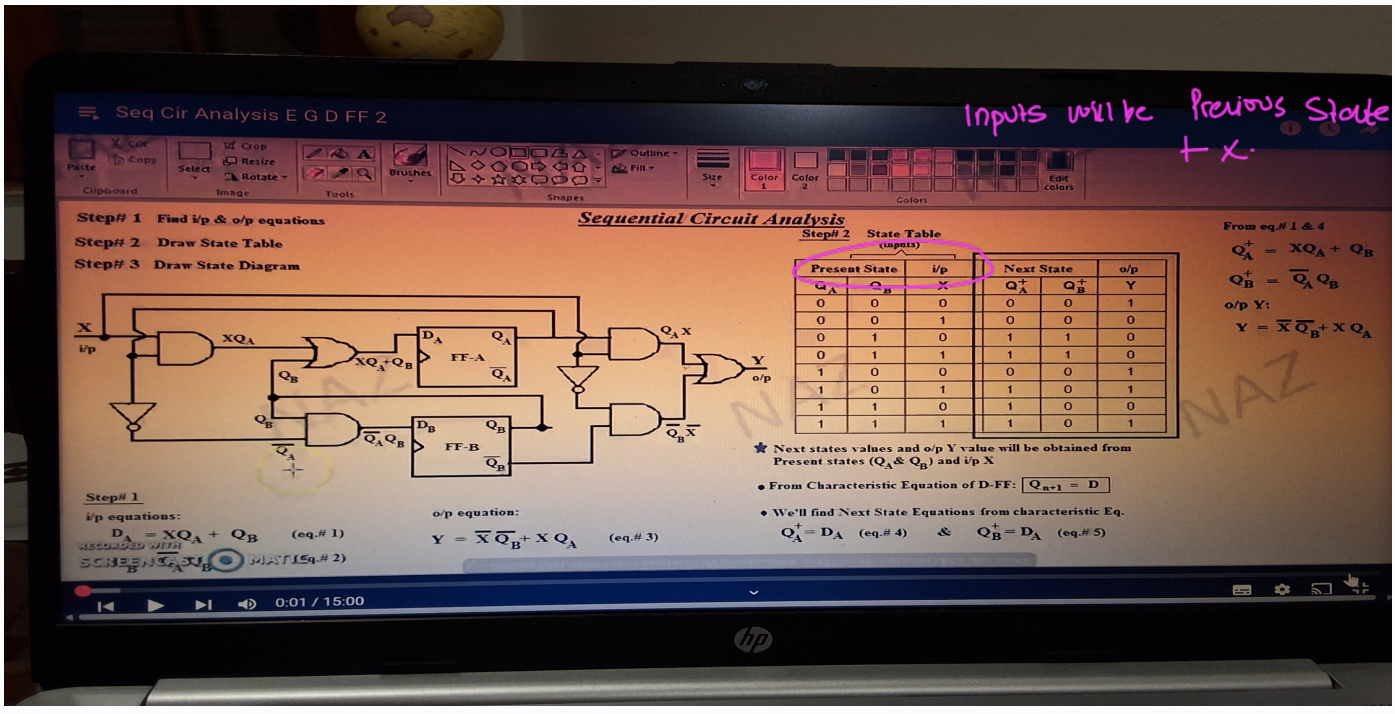
Step 3 Draw State diagram



$D_A = XQ_A + \overline{Q_B}$ $D_B = \overline{Q_A}\,Q_B$

$Y = \overline{X}\,\overline{Q_B} + X Q_A$

Step 2 Making State table



3 inputs variables hence $2^3 = 8$ combinations.

next state values and output value will be obtained from previous states and inputs.

Equation of D ff. characteristic equation $D = Q_{n+1}$

ff input is the output of next state

we find next state equation from characteristic equations

$D_A = Q_A^+$

$D_B = Q_B^+$

$$Q_A^+ = x Q_A + Q_B$$

$$Q_B^+ = \bar{x} Q_B$$

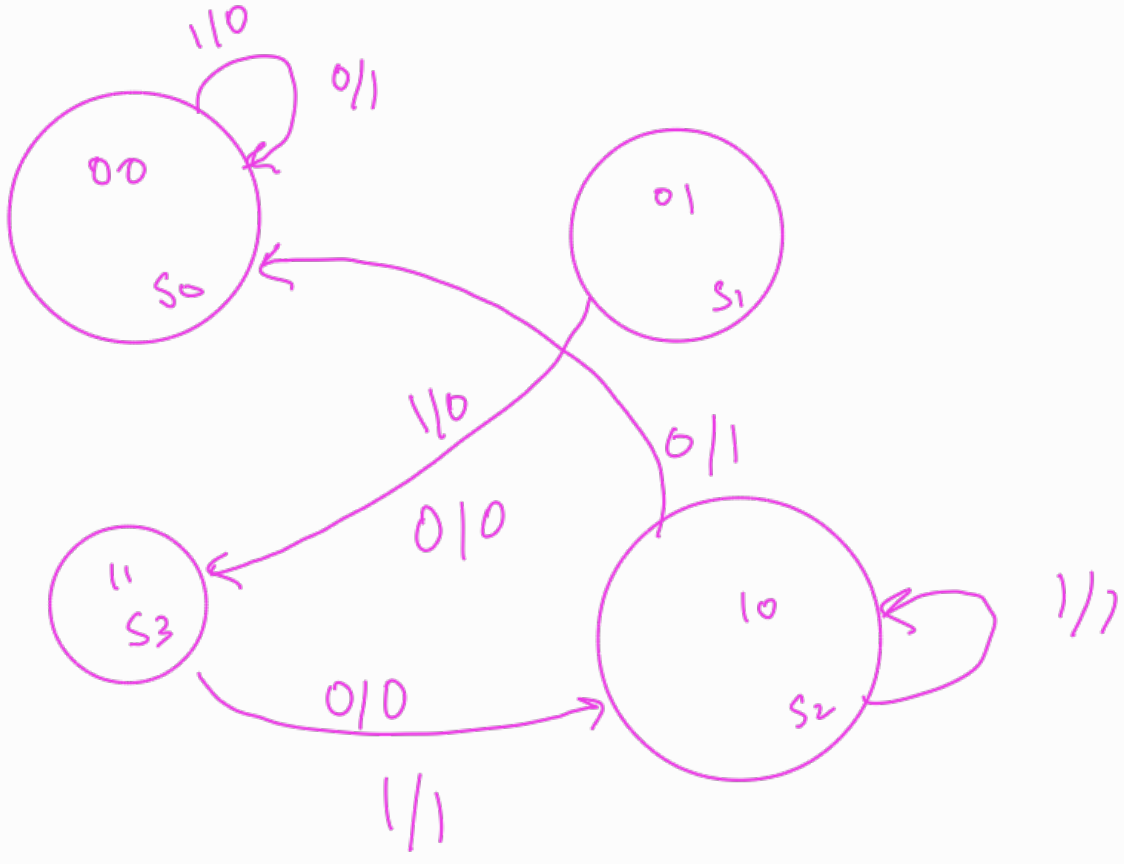
$$Y = \bar{x}\,\bar{Q_B} + x Q_A$$

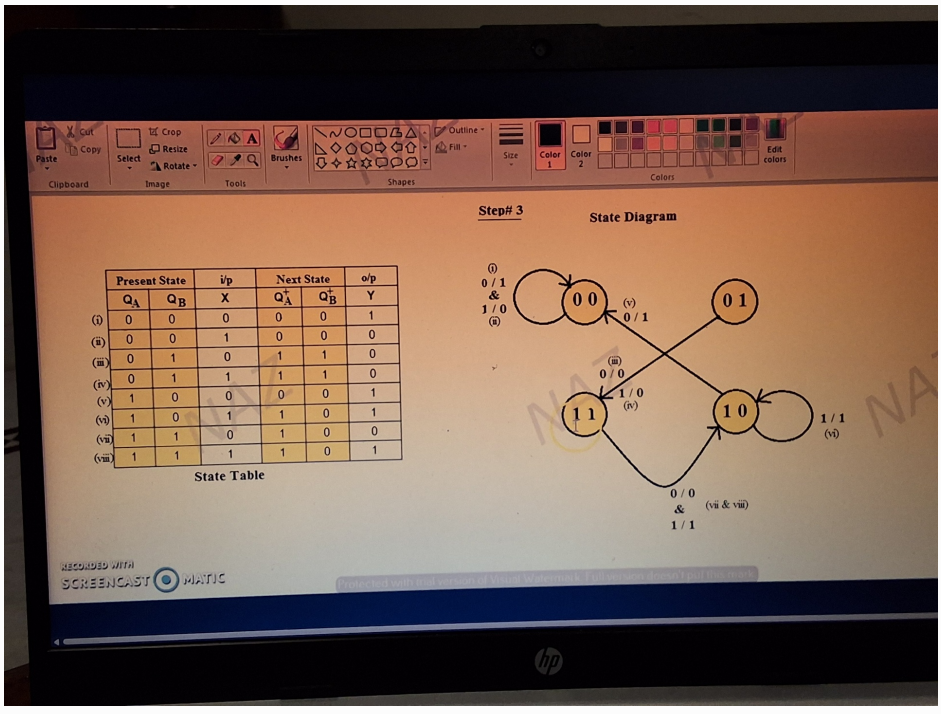
formala for next state Q_A+ and Q_{B+}

now we use these to fill Q_A^+ and Q_B^+

State diagram

$2^2 = 4$ states





Step# 3 **State Diagram**

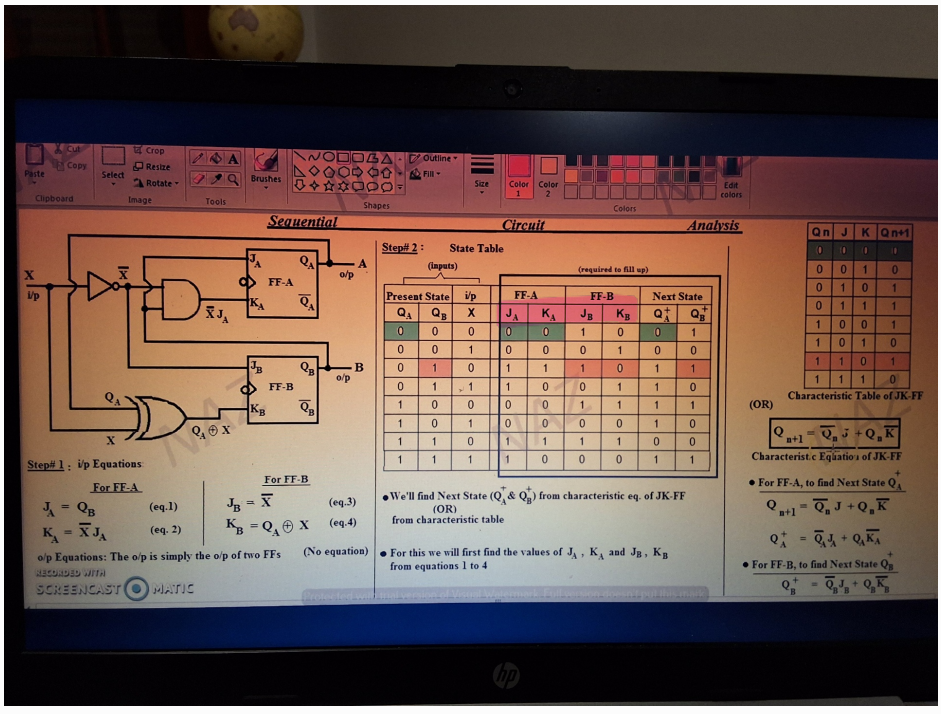
	Present State Q_A	Present State Q_B	i/p X	Next State Q_A^+	Next State Q_B^+	o/p Y
(i)	0	0	0	0	0	1
(ii)	0	0	1	0	0	0
(iii)	0	1	0	1	1	0
(iv)	0	1	1	1	1	0
(v)	1	0	0	0	0	1
(vi)	1	0	1	1	0	1
(vii)	1	1	0	1	0	0
(viii)	1	1	1	1	0	1

State Table

RECORDED WITH SCREENCAST MATIC

JK ff Sequential Analysis

note:- for this we will also include J_A K_A and J_B K_B columns. unlike for the D FF



Sequential Circuit Analysis

Step# 1 : i/p Equations

For FF-A

$J_A = Q_B$ (eq.1)

$K_A = \overline{X}J_A$ (eq. 2)

For FF-B

$J_B = \overline{X}$ (eq.3)

$K_B = Q_A \oplus X$ (eq.4)

o/p Equations: The o/p is simply the o/p of two FFs (No equation)

Step# 2 : State Table

(inputs) (required to fill up)

Present State Q_A	Present State Q_B	i/p X	FF-A J_A	FF-A K_A	FF-B J_B	FF-B K_B	Next State Q_A^+	Next State Q_B^+
0	0	0	0	0	1	0	0	1
0	0	1	0	0	0	1	0	0
0	1	0	1	1	1	0	1	1
0	1	1	1	0	0	1	1	0
1	0	0	0	0	1	1	1	1
1	0	1	0	0	0	0	1	0
1	1	0	1	1	1	1	0	0
1	1	1	1	0	0	1	1	1

- We'll find Next State (Q_A^+ & Q_B^+) from characteristic eq. of JK-FF (OR) from characteristic table
- For this we will first find the values of J_A, K_A and J_B, K_B from equations 1 to 4

Qn	J	K	Qn+1
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

Characteristic Table of JK-FF

(OR)

$Q_{n+1} = \overline{Q_n}J + Q_n\overline{K}$

Characteristic Equation of JK-FF

- For FF-A, to find Next State Q_A^+

$Q_{n+1} = \overline{Q_n}J + Q_n\overline{K}$

$Q_A^+ = \overline{Q_A}J_A + Q_A\overline{K_A}$

- For FF-B, to find Next State Q_B^+

$Q_B^+ = \overline{Q_B}J_B + Q_B\overline{K_B}$

RECORDED WITH SCREENCAST MATIC

State diagram

$2^2 = 4$ states.

Sequential circuit designing

Step 1:. Draw State diagram

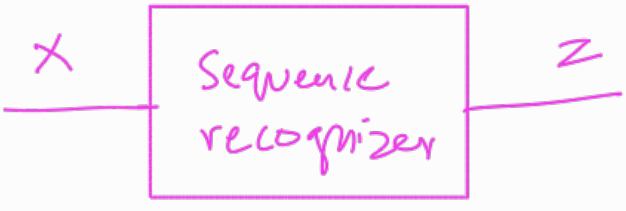
2:. Make State table

3:. FF Input equations

4:. Output equations

5:. Draw logic diagram

e.g



Recognise 101

X:- 0 0 0 1 1 1 0 1 0 1 0 1 1 1 0 1 0 1 0 1 0 1 1 0 1

↳ Input

one bit comming every one cycle

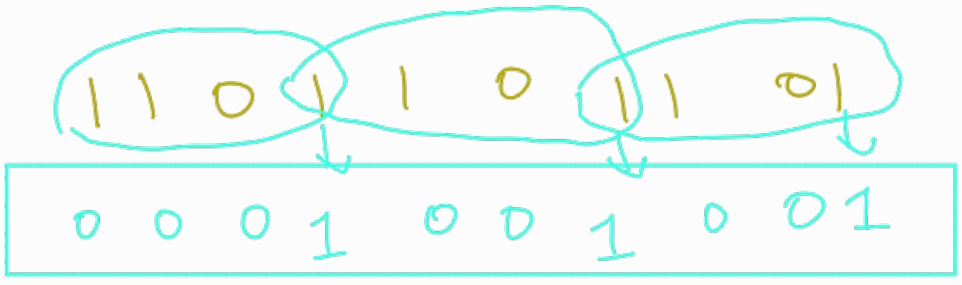
output Z=1 (when pattern found)

Z=0 (when not found)

Question : Detech Sequence "1101"

State diagram 4 bits in Sequences → 4 States

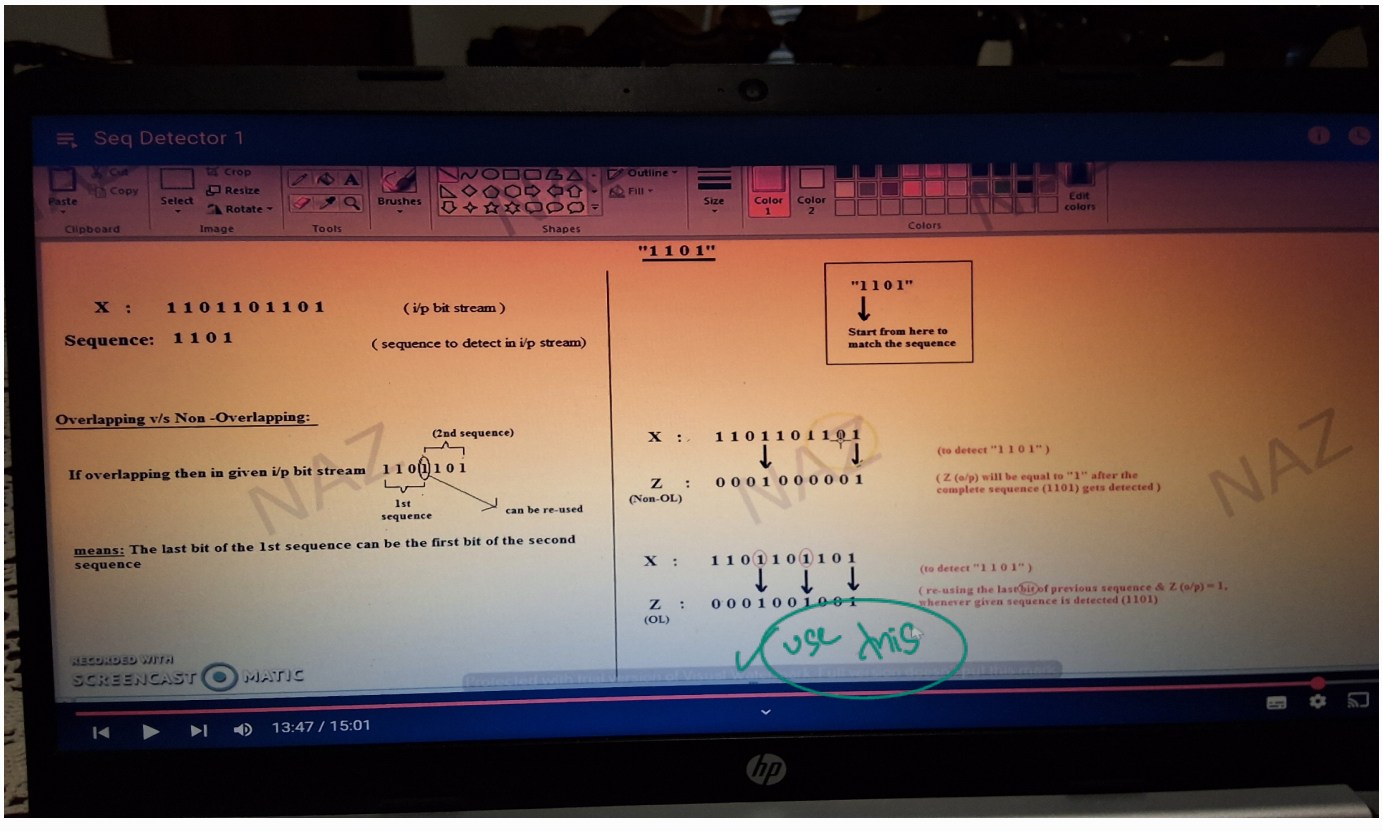
A B C D



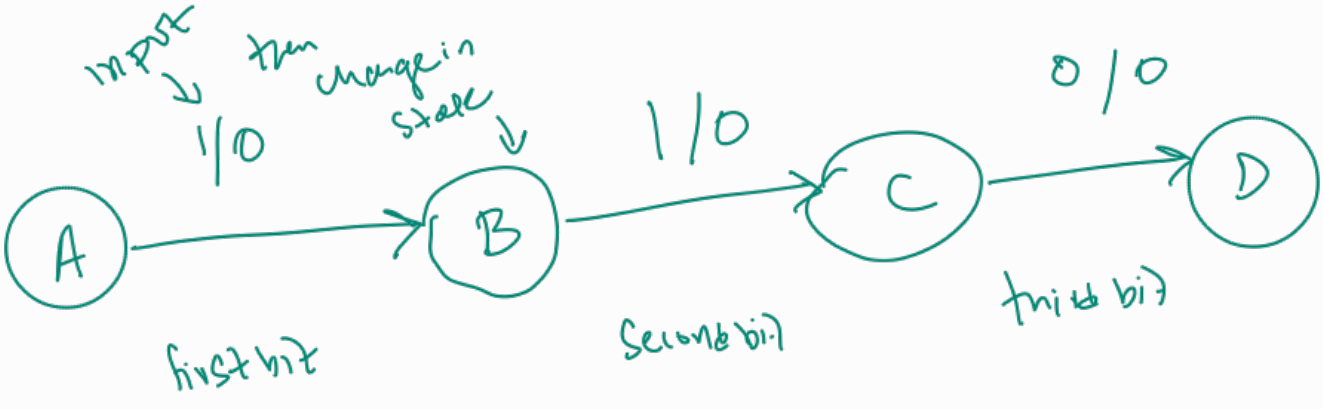
overlapping case

non- Overlapping case

detect 1101



whenever the bit (1101) given is recieved as input is detected we move to another State.



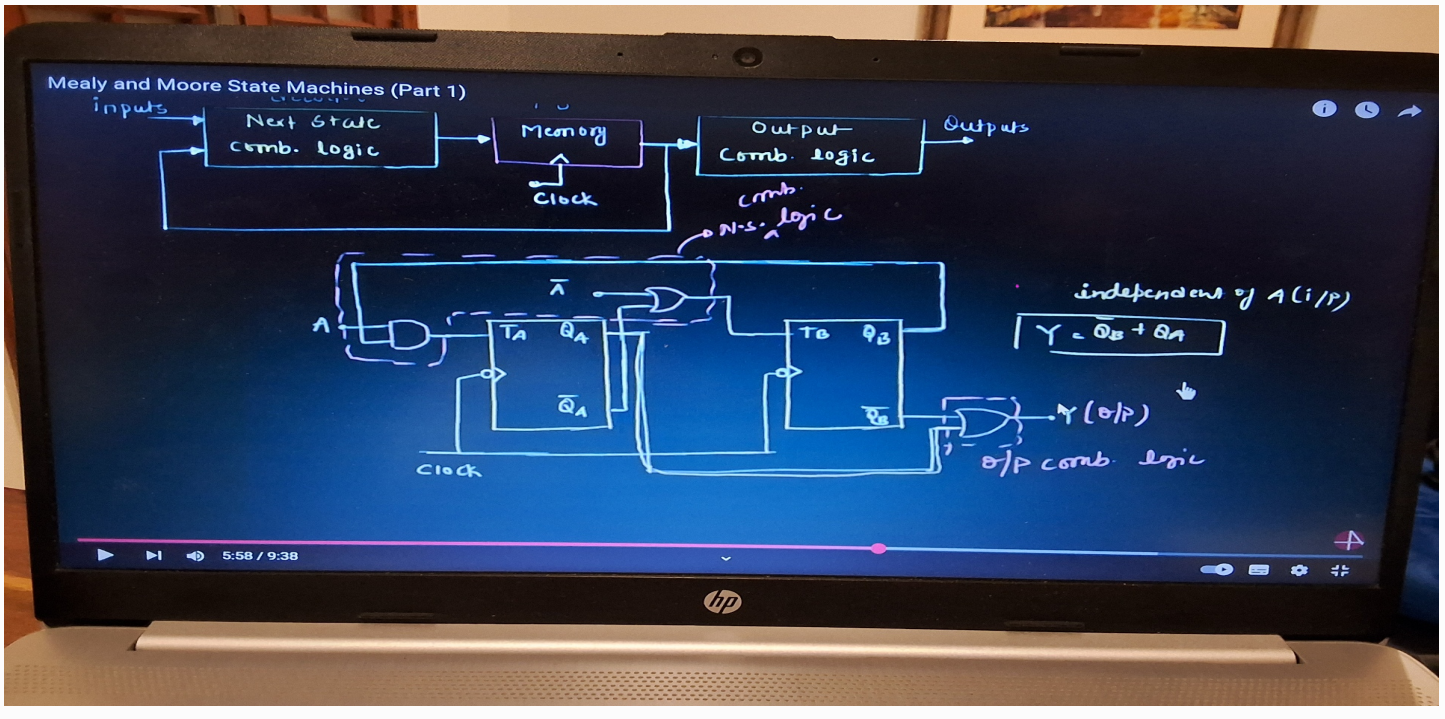
Moore State Machine and Mealy State Machine.

The difference b/w these two circuits is how the output is generated.

Moore → o/p is function of present state only

Mealy → o/p is function of present state + input

Moore State machine

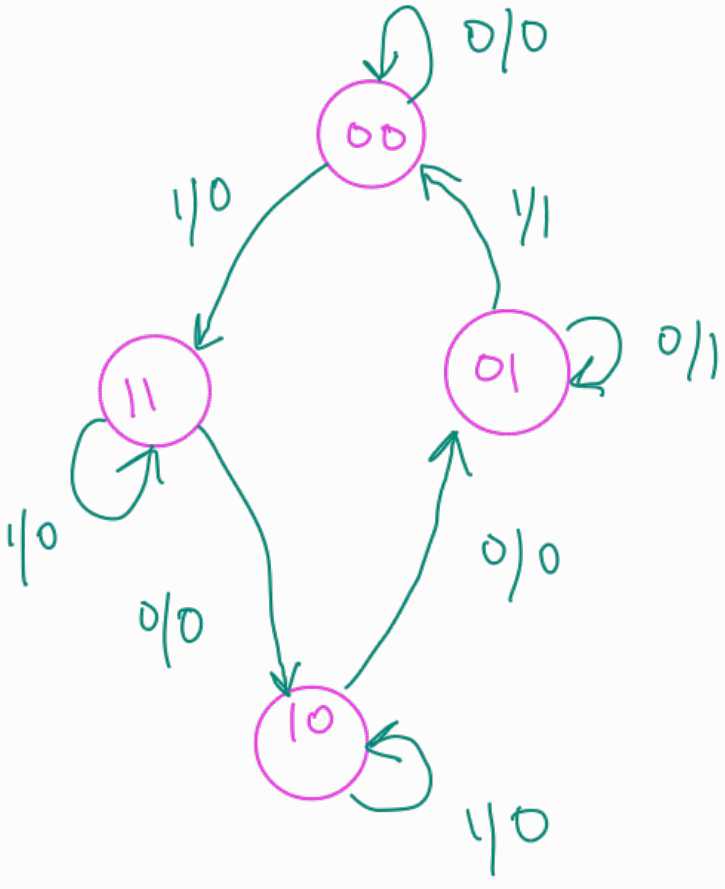


$$Y = \overline{Q_B} + Q_A$$

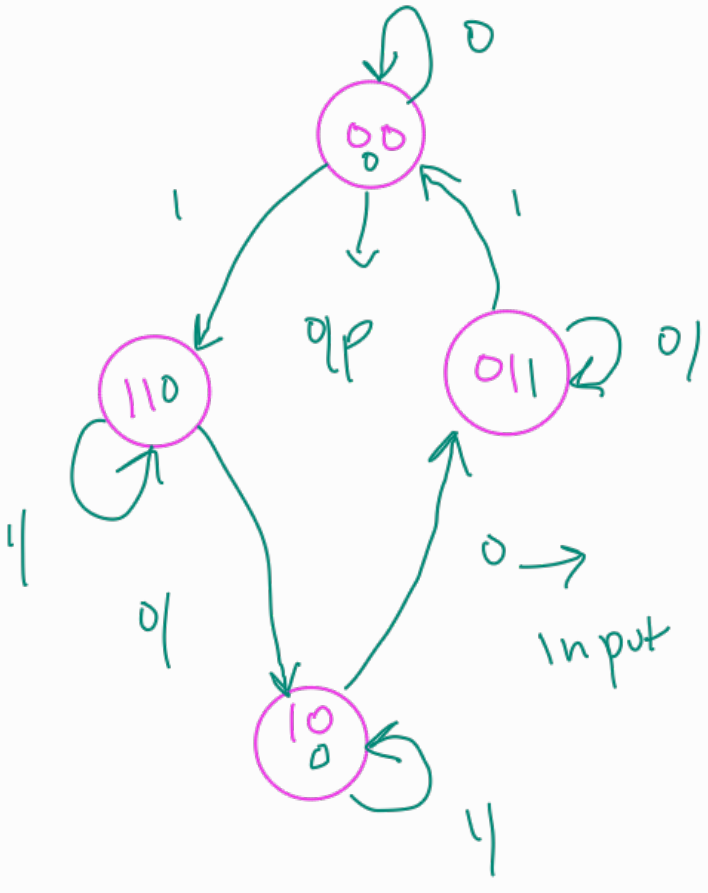
Again this output eq shows it is independent of input hence Moore.

How to convert State diagram to Moore State diagram.

e.g

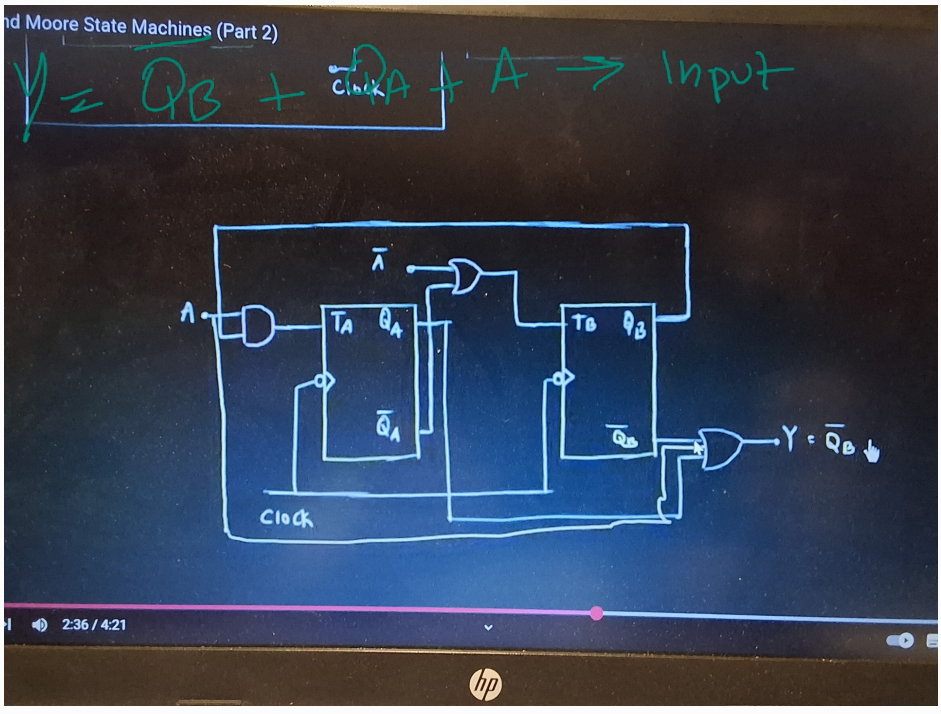


→ write the outputs in the circle.



O/P

Mealy State machine depends on present state + Input



In meany the no of state are less as compared to Moore

In meal for the same stak o/p changes with input.

In moor for the same stak o/p does not change with the input.

State Assignment and reduction.

if we have 3 states e.g
a = 00
b = 01
c = 10
} Assign meant

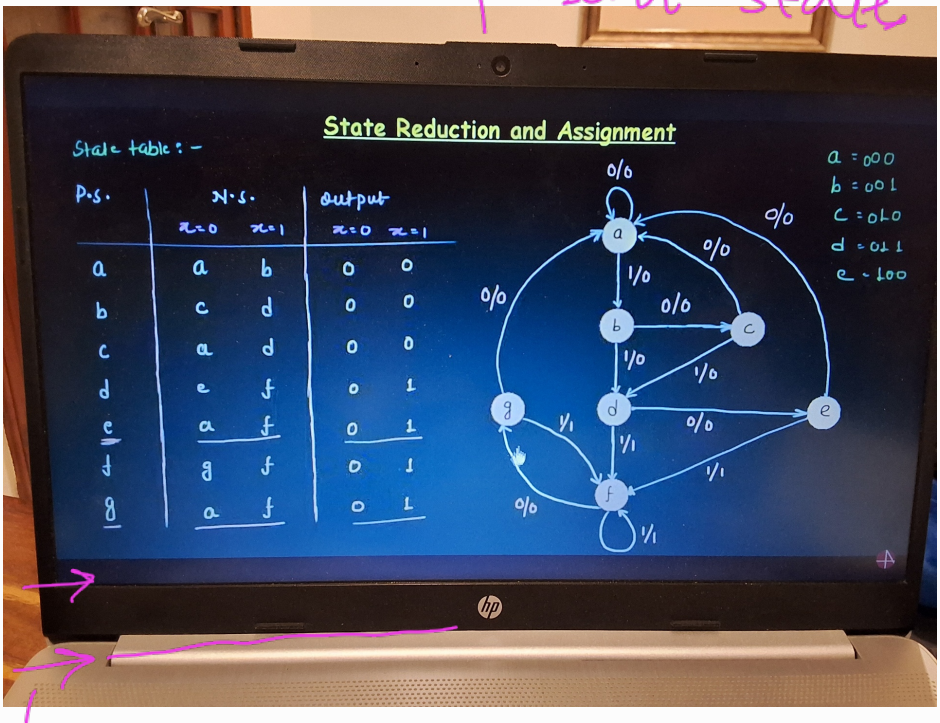
if we have 5 states
use three bits
a = 000
b = 001
c = 010
d = 100
e = 101

State Reduction → to ensure we dont have any unnecessary state / redundant states.

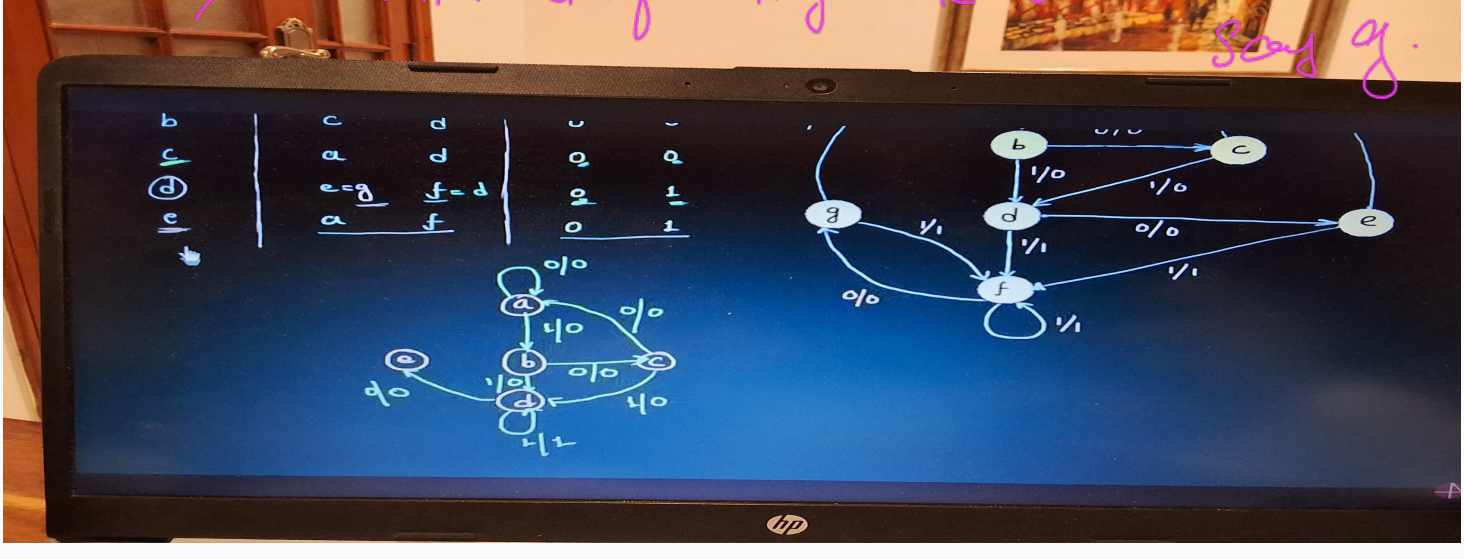
If

(next state and output is same) → Remove it.

for two present state



→ Eliminating any one of them lets say g.



e=g because they are same.

PS	Next State (Inputs: X=0)	Next State (Inputs: X=1)	Output X=0	Output X=1
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	f	0	1
e	a	f	0	1
f	g	f	0	1
g	a	f	0	1

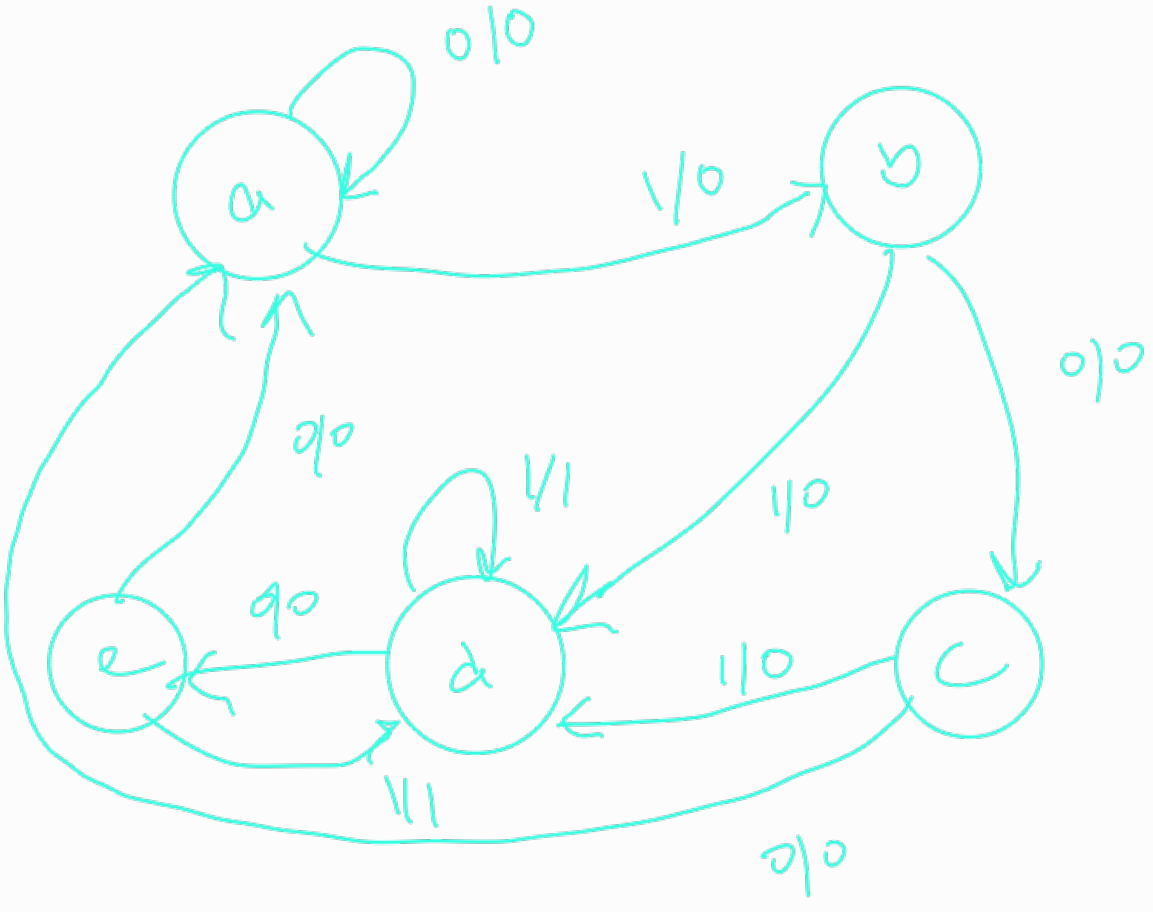
c = g

PS	Next State (Inputs: X=0)	Next State (Inputs: X=1)	Output X=0	Output X=1
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	f	0	1
e	a	f	0	1
f	e	f	0	1

d = f

PS	Next State		Output	
	Inputs: X=0	X=1	X=0	X=1
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	f ~~2~~	0	1
e	a	f ~~2~~	0	1

Max Simplified.



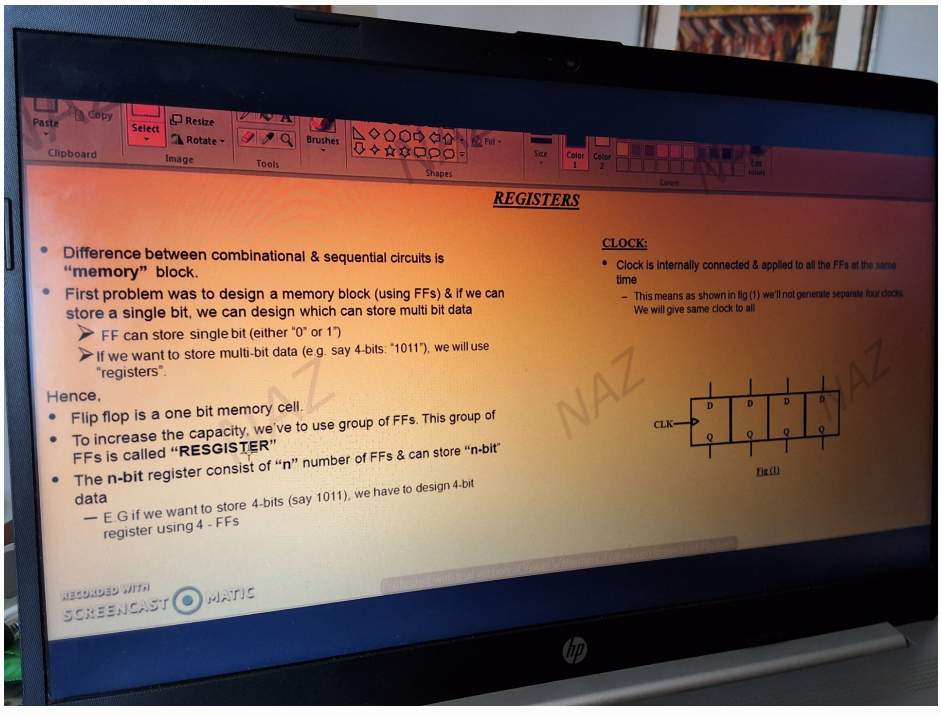
Registers

Diff b/w combinational circuits and sequential circuits is Memory block

FF till now can store Singk bit.

But what if we need to store multiple bits we will use registers (e.g 1101)

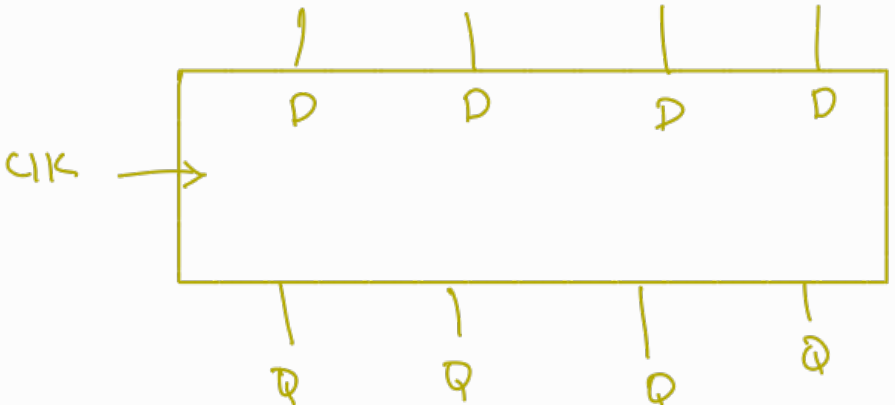
with the help of FF we create Registers.



The n bit register consist of n bit ff and stores n bit data

if we want to store (1101) we will use 4 bit Register using 4 FF.

clock is same for all T-F

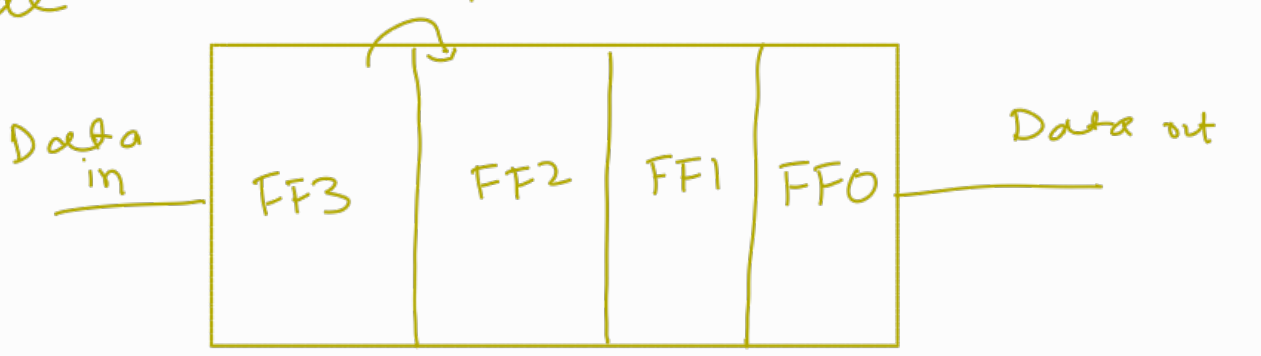


Trigger applied to all simultaneously

e.g store 1011

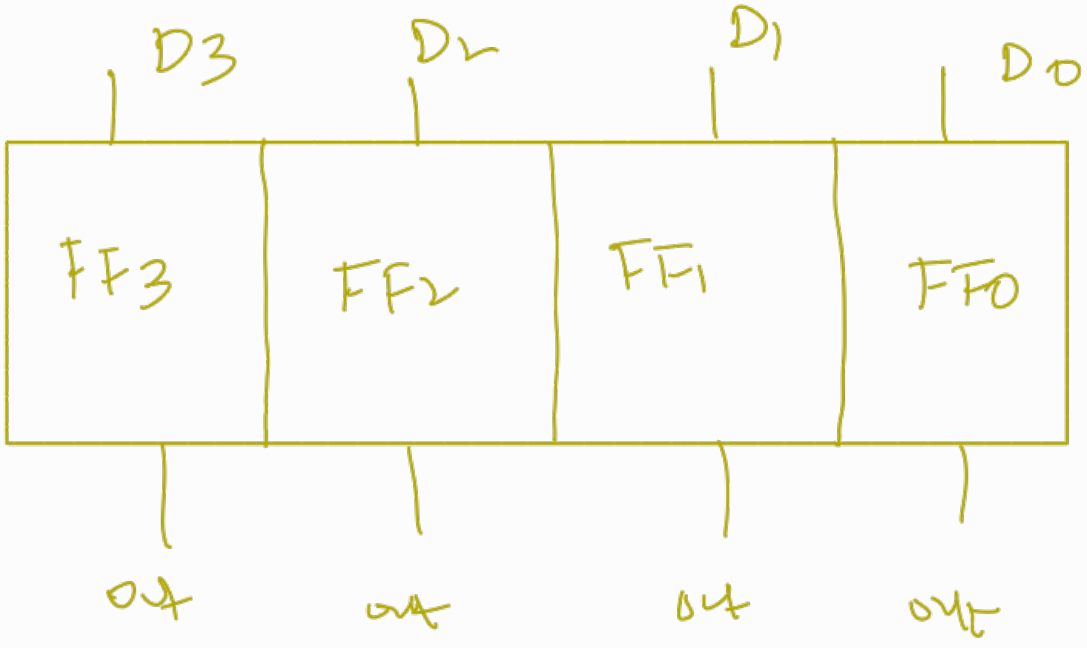
Types of data format.

i) Serial

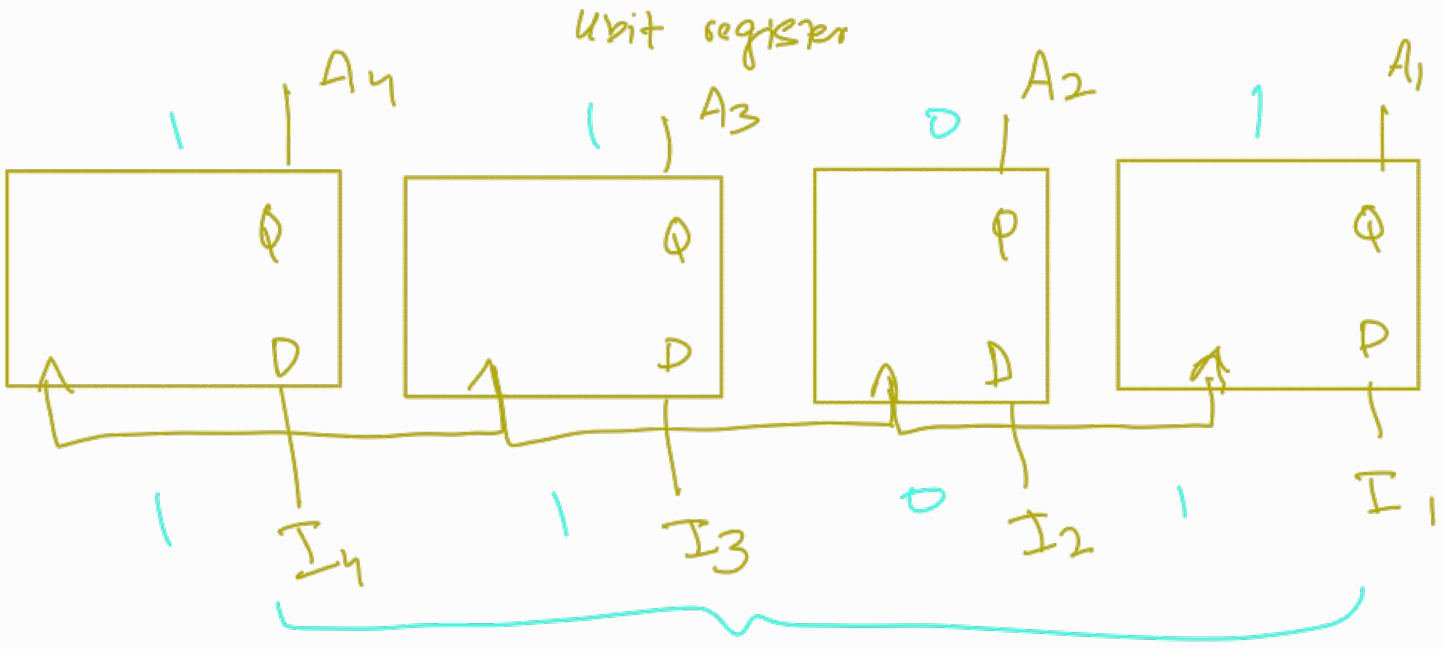


data inputs one by one and we get output one by one.

ii) Parallel form.



In addition to FF, a register may have combinational circuits gates that perform data Processing tasks. The gates control when and how new information is transfered into the register.



lets say input is 1101 then for DFF it sends the same input as output.

↳ when clock is positive edge triggered

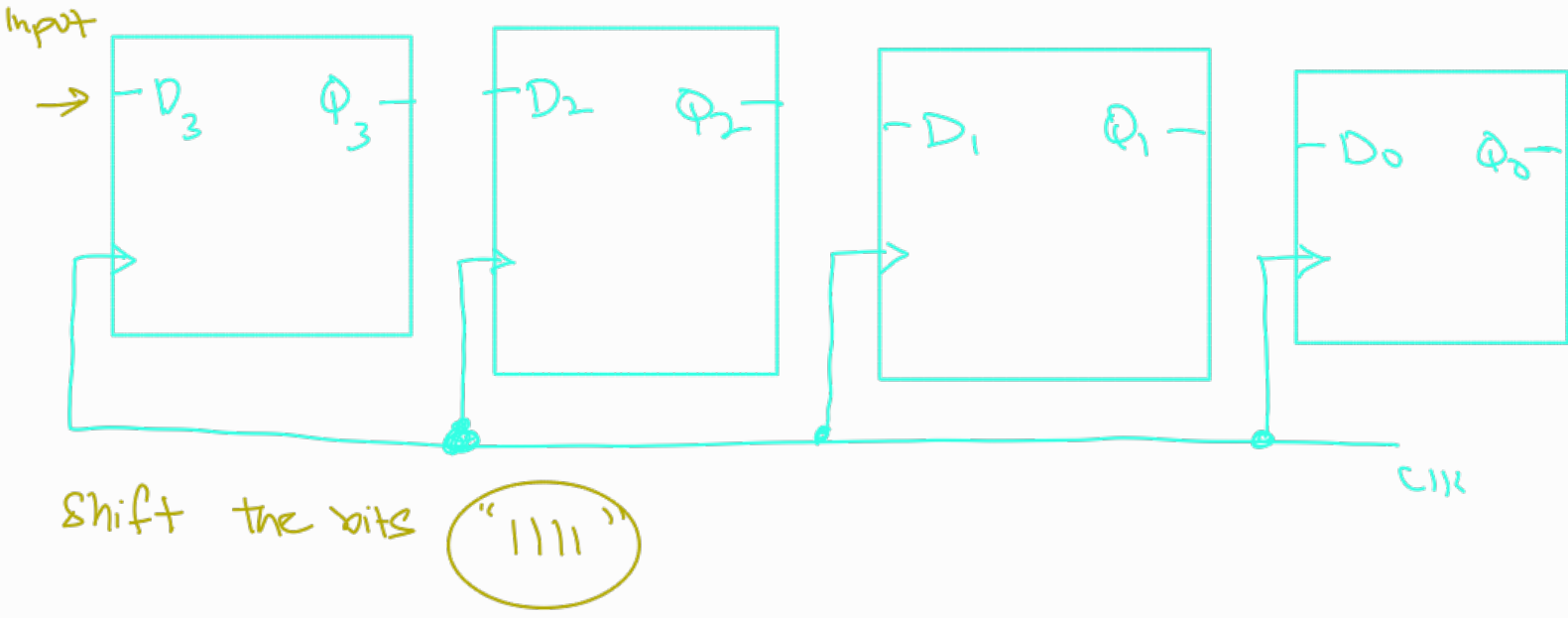
when clock is negative edge triggered

Dff holds the previous state. and it will not change its output even with new inputs.

Shift register:.

if we want data to be shifted from FF-3 to FF-2 then from FF-2 to FF1 and then to FF1 to FF0.

e.g



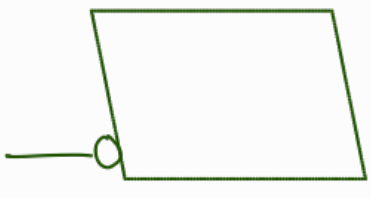
Shift the bits "1111"

note:. Logic 1 → means simply on.

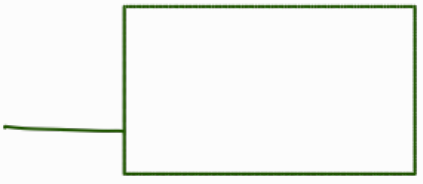
Logic 0 → means simply off.

Active High means machine gets turned on from logic 1.

Active low means machine gets turned on from logic 0. (Inverter)

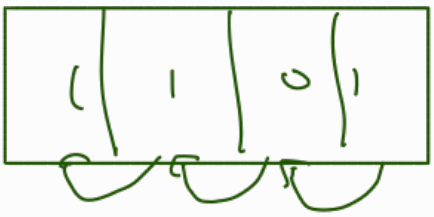


on at Logic 0



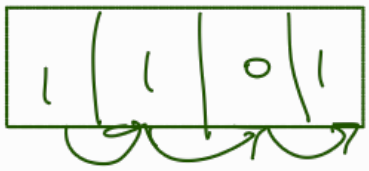
on at logic 1

Shift Register:: Left shift



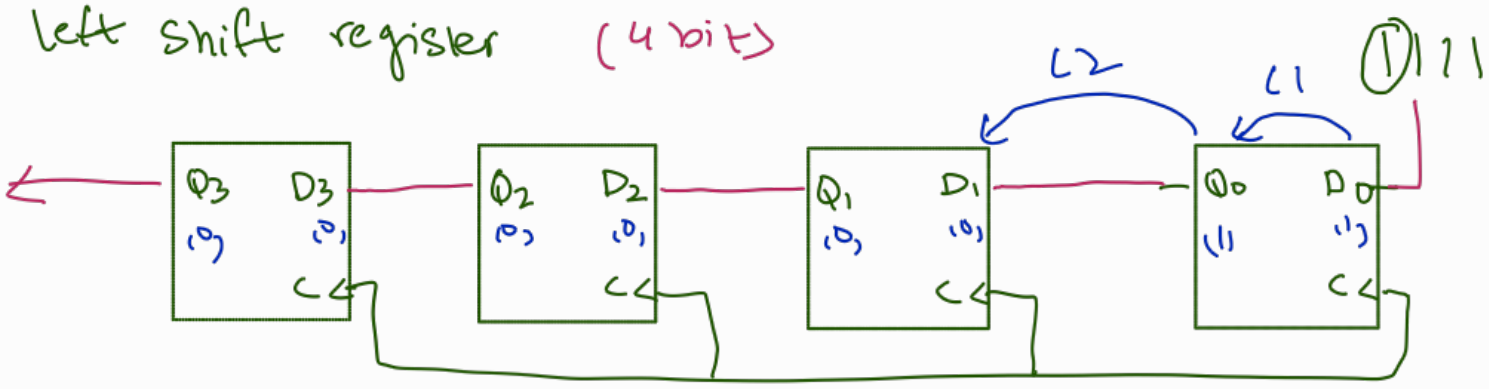
Start from LSB

Right shift



Start from MSB

Left shift register (4 bit)



coming out from leftside

neg edge triggered will generate output only then.

Note:: n bit Input → n bit number of clocks

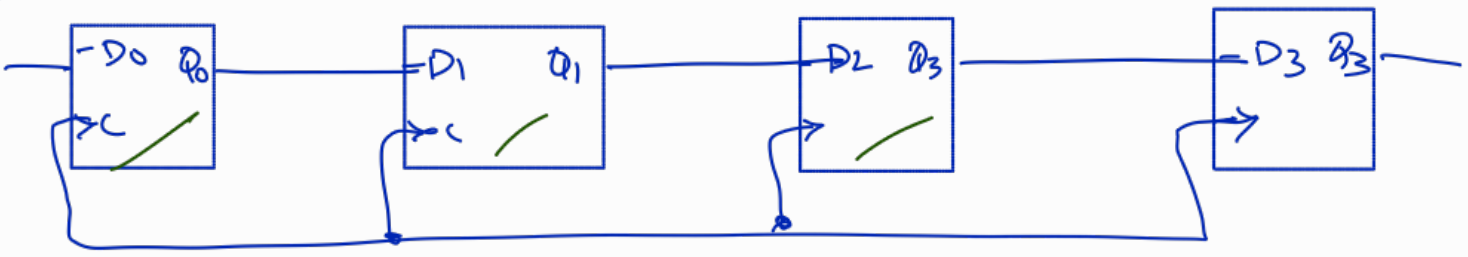
	D_3	D_2	D_1	D_0	Q_3	Q_2	Q_1	Q_0
	0	0	0	0	0	0	0	0 ← assume
C1	0	0	0	1	0	0	0	1
C2	0	0	1	1	0	0	1	1
C3	0	1	1	1	0	1	1	1
C4	1	1	1	1	1	1	1	1

D_3	D_2	D_1	D_0	Q_3	Q_2	Q_1	Q_0 (Assume)
0	0	0	0	0	0	0	0 (Initially)
0	0	0	1	0	0	0	1
0	0	1	1	0	0	1	1

next bit

Right Shift register:

1111

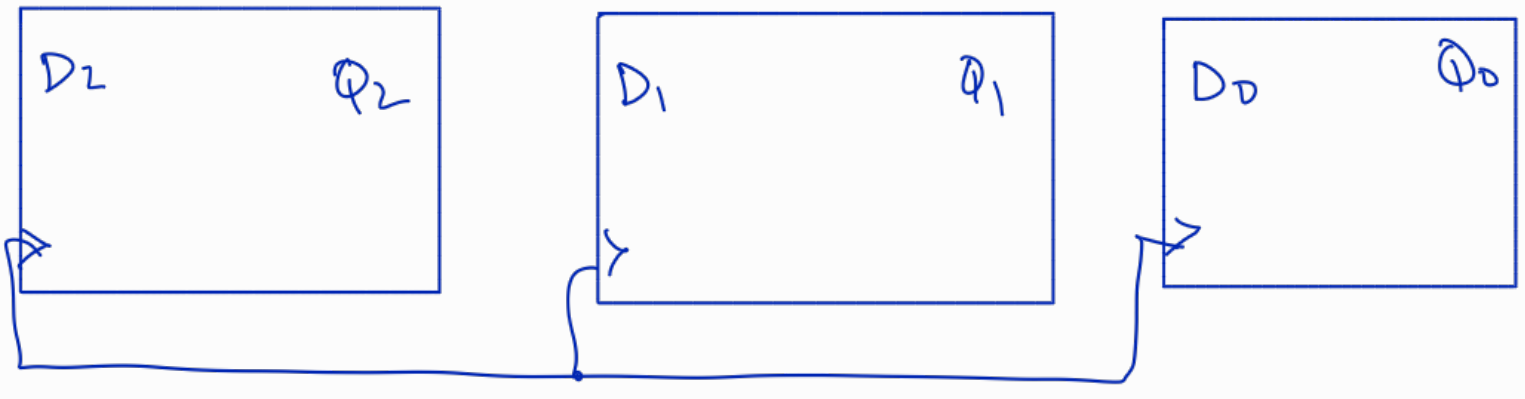


	D_3	D_2	D_1	D_0	Q_3	Q_2	Q_1	Q_0
	0	0	0	0	0	0	0	0
C_1	0	0	0	1	0	0	0	1
C_2	0	0	1	1	0	0	1	1
C_3	0	1	1	1	0	1	1	1
C_4	1	1	1	1	1	1	1	1

At C3 three FF are actived only

Bidirectional Shift Register:: → Shifts bits in both directions

Considering a 3 bit number



Storage Registers: Just to store data and use it when needed. Best example PIPO.